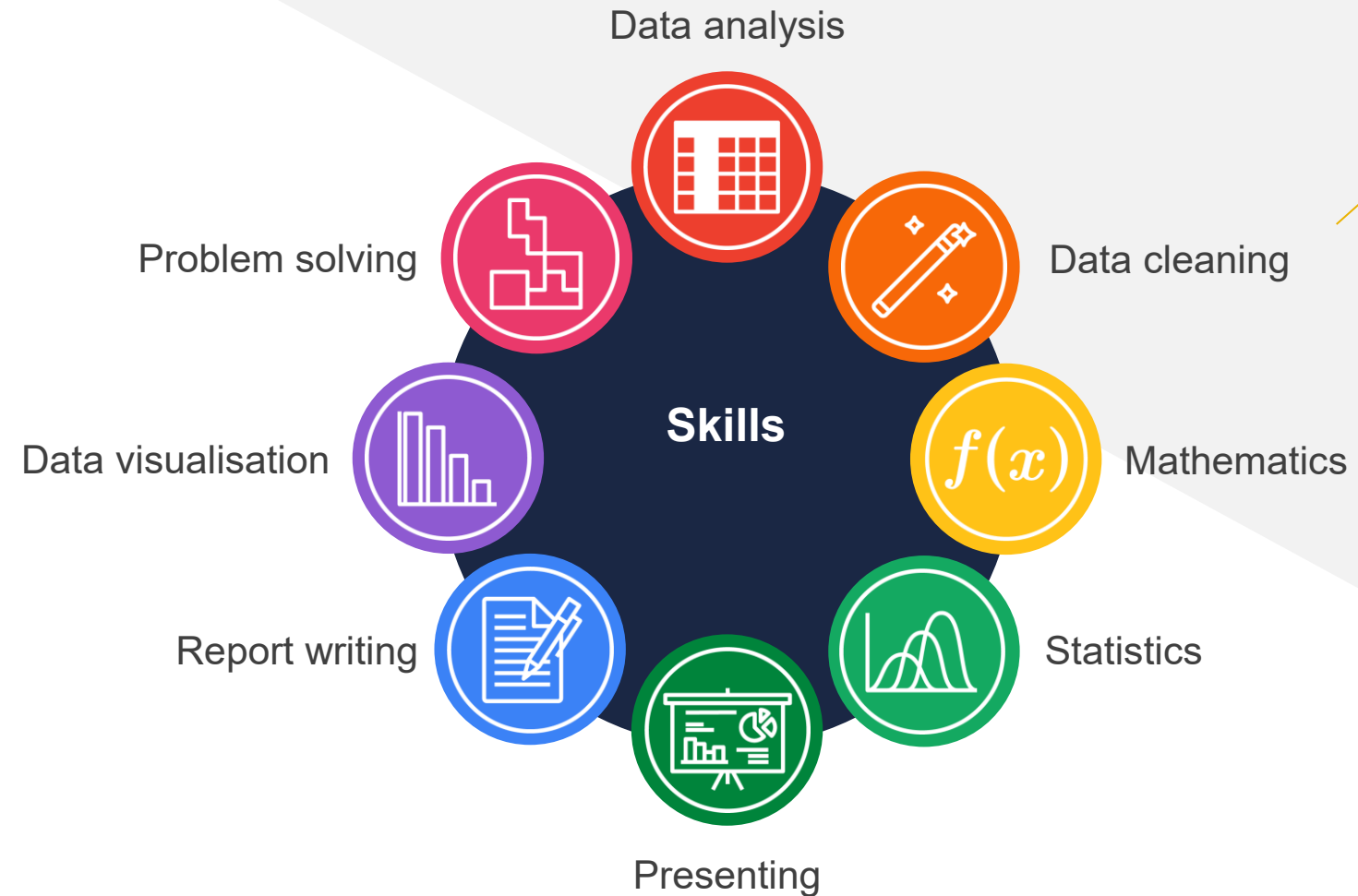




Introduction

Content



Agenda

- Data Analysis Definition
- Data Analysis VS Data Analytics
- Data Analysis Applications
- Real World Business Cases

What is Data Analysis?

What is data?

Data are **collections of distinct values** represented by characters that are stored in or used by a computer.



Letter



Digit



Special character

A **character**, abbreviated as **char**, is a single object used to represent text, numbers, and symbols.

Examples of data:

Electrical, 1018445016, no power, -47072, -3771400, 15.07.2022

69 108 101 99 116 114 105 99 97 108 44 32 49 48 49 56 52 52 53

01000101 01101100 01100101 01100011 01110100 01110010 01101001
01100011 01100001 01101100 00101100 00100000

What is data?

Data can be of **types** beyond a single character.



Boolean



Text



Number

Boolean (logical true or false), text (strings), and numbers (float or integer) are commonly stored in **tabular format**. In greater quantities, these types are stored in database tables.



Image



Sound



Video

Image, sound, and video files are **more complex to store and analyze** in tabular format. They can be represented in a table through the storage path of the data file.

What is a table?

A data **table** is a grid with data arranged in labeled rows and columns.

Column names	Division	Location	Billing Class	Amount	Quantity	Number of contract accounts
Row 1	ELECTRICITY	ACACIA PARK	Commercial customers	R3 710,700	1 000,000 KVA	1
Row 2	ELECTRICITY	AMANDEL RUG	Domestic customers	R1 927,420	768 000,000 W	1
Row 3	ELECTRICITY	ATHLONE	Commercial customers	R49 252,220	22 722,594 KWH	4
Row 4	ELECTRICITY	ATHLONE	Departmental	R8 329,520	0,924 KWH	5
	ELECTRICITY	BELLVILLE CBD	Departmental	R174 840,060		2
	ELECTRICITY	BURGUNDY	Not assigned	R83 140,040	44 586,560 KWH	1
	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6

“... The goal is to turn data into information, and information into insight.”

- Carly Fiorina (former CEO at HP)

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0
...	...	...	...	...	...	...	...	...	...	...	...	...
913	45	M	TA	110	264	0	Normal	132	N	1.2	Flat	1
914	68	M	ASY	144	193	1	Normal	141	N	3.4	Flat	1
915	57	M	ASY	130	131	0	Normal	115	Y	1.2	Flat	1
916	57	F	ATA	130	236	0	LVH	174	N	0.0	Flat	1
917	38	M	NAP	138	175		Normal	173	N	0.0	Up	0

	count	mean	std	min	25%	50%	75%	max
Age	918.0	53.510893	9.432617	28.0	47.00	54.0	60.0	77.0
RestingBP	918.0	132.396514	18.514154	0.0	120.00	130.0	140.0	200.0
Cholesterol	918.0	198.799564	109.384145	0.0	173.25	223.0	267.0	603.0
FastingBS	918.0	0.233115	0.423046	0.0	0.00	0.0	0.0	1.0
MaxHR	918.0	136.809368	25.460334	60.0	120.00	138.0	156.0	202.0
Oldpeak	918.0	0.887364	1.066570	-2.6	0.00	0.6	1.5	6.2

Data versus information

Data

- Generally **raw**, **unorganized**, and **unfiltered**.
- Comprise of all **elements or facts required** for analysis.
- **Independent** of information.
- **Meaningless** on its own.
- **Not sufficient for decision-making**.

Examples:

A book contains **words** (data) and as one reads the book to completion, one **infers** what the book is about (information).

Every day a cashier **records sales items** (data). At the end of the month, the manager looks at the total number of sales and deduces if they made **a profit or loss** (information).

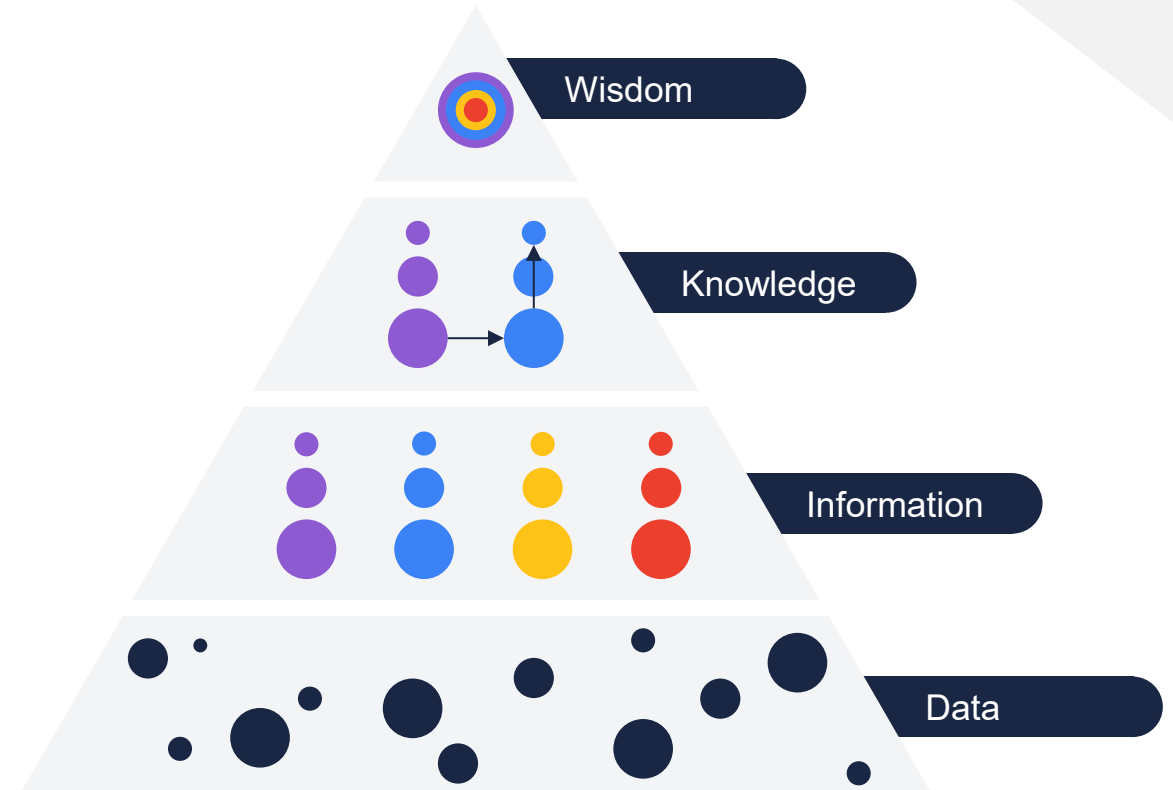
Information

- **Refined, structured, and evolved** from raw data.
- Puts facts or elements into **context**.
- **Dependent** on data.
- **Meaningful**; derived from analysed and interpreted data.
- **Sufficient for decision-making**.

The DIKW pyramid

The DIKW pyramid represents the **relationship** between **data**, **information**, **knowledge**, and **wisdom**.

- **Wisdom** is the ability to make **sound judgements** and **decisions** based on experience, **knowledge**, and understanding.
- **Knowledge** is the **understanding** or **comprehension** of the **information** we derive from data.
- **Information** is **processed** and **organized** data presented in a meaningful context.
- **Data** are **raw**, **unprocessed**, and **unorganised** facts that are seemingly random and useless until they're processed and organised.



What are data used for?

Decision-making

Data are used to make more informed decisions.

Product development

Data are used to build and develop products and services for the improvement of our lives or businesses.

Storytelling

Data are used to assist us in better contextualising the stories we tell.

Data-driven decision-making is using metrics and facts derived from data to guide strategic decisions that align with the business goals, objectives, and KPIs (Key Performance Indicators).

Data can potentially add value by providing **insightful information** that **enhances our decision-making abilities**.

"Without data, you're just another person with an opinion."

Some examples of **real-world uses** include prediction of future sales, purchasing behaviours, fraud detection products, marketing campaign analysis, and improving customer acquisition and retention.

There is no analysis without data

DATA is at the heart of **data analysis**.



Data analysis involves examining data to answer questions, identify trends, and extract insights.



DATA ANALYSIS is all about uncovering useful insights and trends from **data**.

Quantitative and qualitative data analyses

Quantitative and qualitative data analyses are important because they enable us to gain a **more comprehensive understanding** of complex phenomena and make **data-driven decisions**.

Quantitative data analysis involves numerical measurement and statistical analysis.

It allows us to **measure and analyse numerical data** using **statistical methods**, enabling us to **identify patterns**, trends, and relationships between variables.

It is useful for making predictions, testing hypotheses, and identifying cause-and-effect relationships.

Qualitative data analysis involves exploring patterns and themes in non-numerical data.

It allows us to **explore and interpret non-numerical data**, such as text, images, or videos.

It is useful for understanding the **context of a problem** and people's attitudes, behaviours, etc.

Both types of analysis are important because they provide **different ways of understanding and interpreting data**.

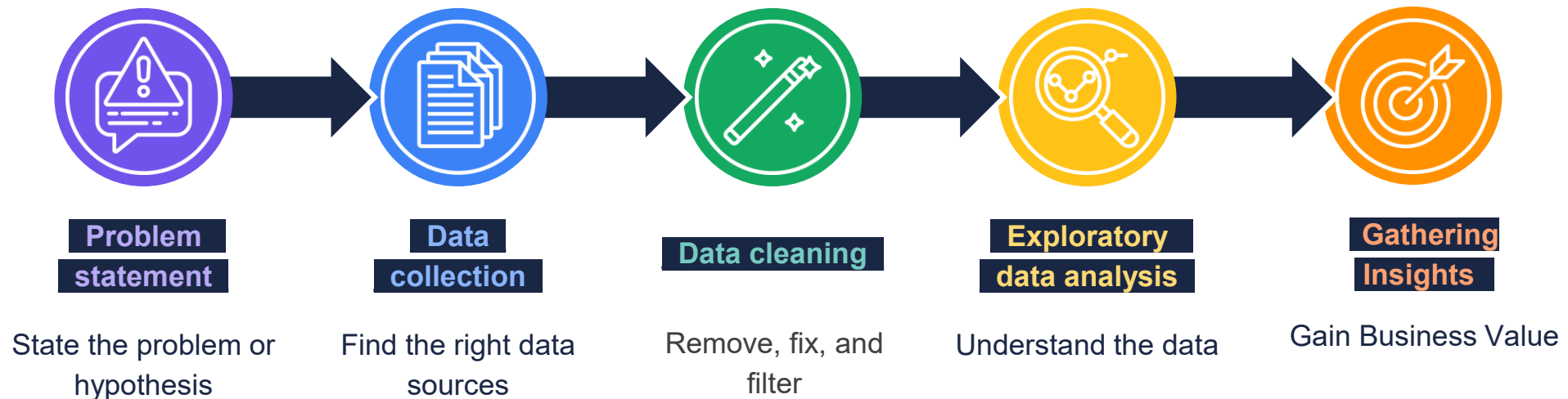
There is no business value without data analysis



When data analytics is used in business,
it's often called **Business Analytics**

The Data Analysis Process

The data analysis process is a systematic approach to **transforming a data problem** into a **data-driven insight**.



This approach to data analysis helps us to **discover meaningful** patterns, relationships, and trends and helps us **make informed decisions** to drive strategic business actions and enhance operational efficiency.

“...nothing is more certain to cause a project to **fail** than a **misunderstanding of the problem** you are solving.”

- Lenny Rachitsky (former developer at Airbnb)

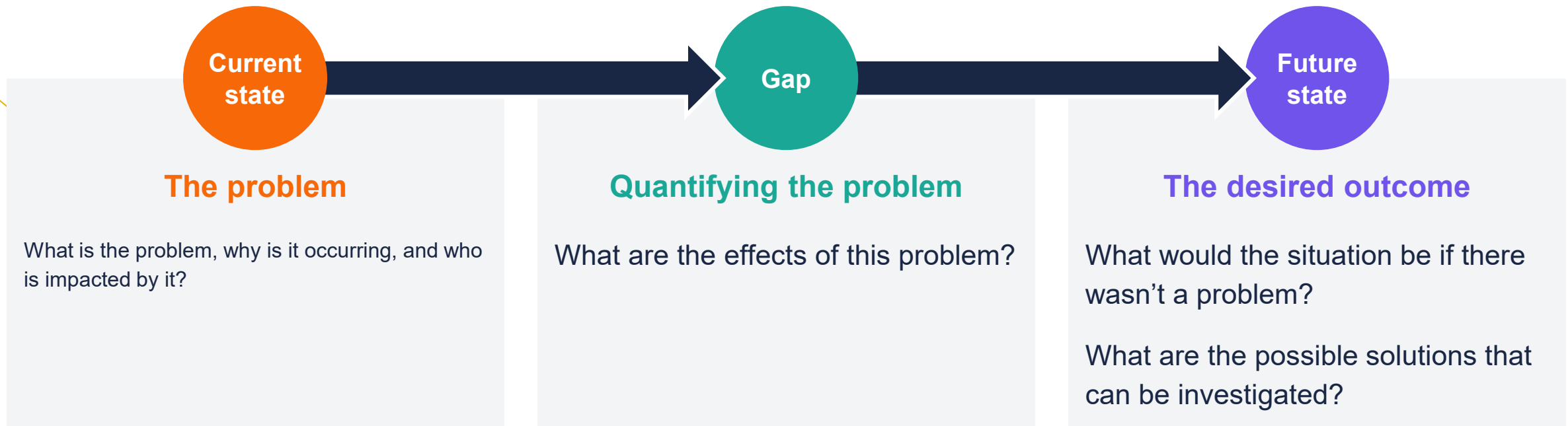
Problem statement



A **problem statement** identifies the gap between the **current (problem) state** and the **desired (outcome) state**.

It helps us **define the scope and objectives** of our analysis and ensures that our insights are **relevant**.

KEY ELEMENTS of a problem statement:



Problem statement

CHARACTERISTICS of a **good** (and not so good)
problem statement:

GOOD...

Specific

Brief and concise

Clear, unbiased, and
measured

NOT SO GOOD...

Generic

Verbose

Open to misinterpretation

How to ask the right questions?

S M A R T

				
S-pecific	M-easurable	A-ction-oriented	R-elevant	T-ime-bound
Is the question specific? Does it address the problem? Does it have context? Will it uncover a lot of the information you need?	Will the question give you answers that you can measure?	Will the answers provide information that helps you devise some type of action plan?	Is the question about the particular problem you are trying to solve?	Are the answers relevant to the specific time being studied?

The 5W2H method

The 5W2H (what, why, where, when, who, how, how much) method enables us to ask the **right questions** in an appropriate order to define an **actionable** problem statement.

Check that your problem statement answers these questions.

WHAT is the problem that needs to be solved?

WHY is it a problem?

Already known explanations contributing to the problem.

WHERE does the problem present itself?

The locations and/or products affected by the problem.

WHEN is the problem observed?

The times when the problem is occurring.

WHO is experiencing the problem?

The stakeholders that are impacted by the problem.

HOW is the problem manifesting?

The symptoms of the problem.

HOW MUCH of a problem is it?

The magnitude, error rate, and trend of the problem.

Give it a try



See if you can recognise the **WHAT** and **WHY** of the **5W2H method** in the following problem statement.

My throat is very sore because I have a cold from sitting outside in the rain last night.

WHERE

WHEN

WHO

HOW

HOW

MUCH

Can you recognise any of the **other five 5W2H** questions in the problem statement?

Give it a try – explanation



See if you can recognise the **WHAT** and **WHY** of the **5W2H method** in the following problem statement.

My throat is very sore because I have a **cold** from **sitting outside in the rain** last night.

“**My**” is the **WHO**, indicating that the person giving the statement is the one experiencing the problem.

The **HOW MUCH** is the qualitative “**very**” in the problem statement.

You might think that the problem was “**throat is sore**” but that is the symptom of the cold, i.e. **HOW** we know that we have a problem.

“**throat is sore**” could also be interpreted as **WHERE** the problem is presenting itself.

We can infer that the **WHEN** is implicitly at the point in time when the statement was given.

We see that some of the questions are often implicitly answered. However, with more **complex statements** we try to be as **clear and explicit** as possible.

More challenging example



Considering the United Nations Sustainable Development Goal (UN SDG) on quality education, see if you can recognise each of the seven questions of the **5W2H method** in the following problem statement.



WHAT

People in developing countries are further divided into social segments due to language differences resulting from varying mediums of education in public and private schools.

WHY

WHERE

WHEN

WHO

HOW

HOW

MUCH

*Refer back to the simpler example to help you in this exercise.

More challenging example



Considering the United Nations Sustainable Development Goal (UN SDG) on quality education, see if you can recognise each of the seven questions of the **5W2H method** in the following problem statement.



WHAT

People in **developing countries** are **further** **divided into social segments** due to **language differences** resulting from **varying mediums of education in public and private schools.**

WHY

WHERE

WHEN

WHO

HOW

HOW

MUCH

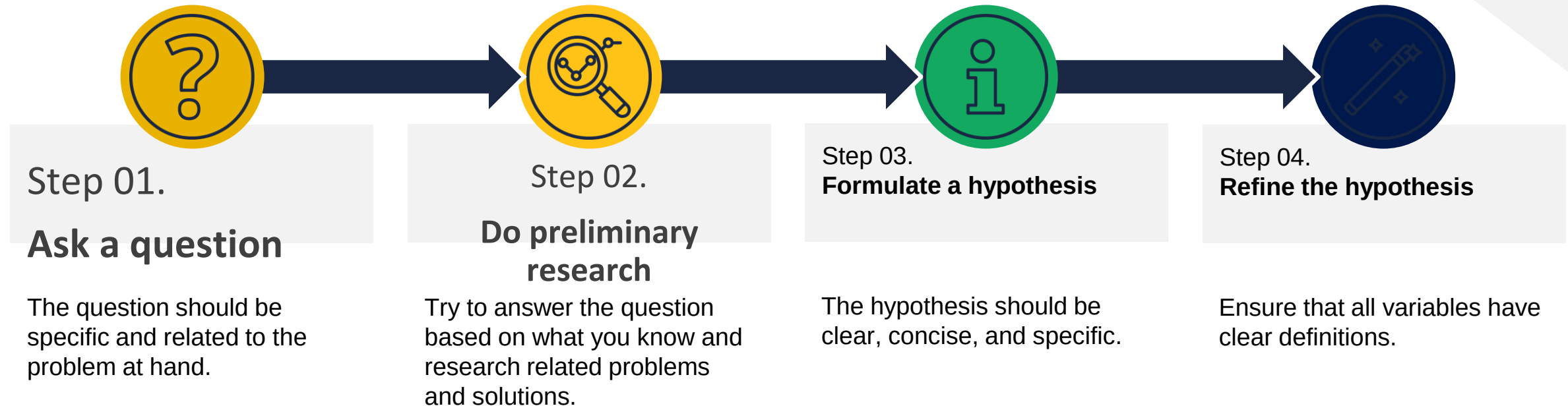
Comparing this problem statement to the cold and symptom example shows the true **WHAT.** Let's rewrite the previous example in the format of this statement:

"... **divided into social segments** due to **language differences** resulting from **varying mediums of education** ..."

"... **throat is sore** due to **the cold** resulting from **sitting outside in the rain** ..."

What is a hypothesis?

A hypothesis is a **proposed cause and effect** for a particular phenomenon or problem which has not yet been proven correct.



Try on your own



See if you can rewrite the following statement into a **hypothesis**.



People in developing countries are further divided into social segments due to varying mediums of education in public and private schools.

Let's first try to identify the **cause** and **effect** for this particular problem (which has not yet been proven true or false).

Try on your own



See if you can rewrite the following statement into a **hypothesis**.



cause

effect

People in developing countries are further **divided into social segments** due to **varying mediums of education in public and private schools**.

A possible hypothesis

Varying mediums of education in public and private schools **do not further divide people** in developing countries into **social segments**.

Another possible hypothesis

Standardising mediums of education across public and private schools in developing countries will **lead to fewer language differences and result in a greater sense of community**.

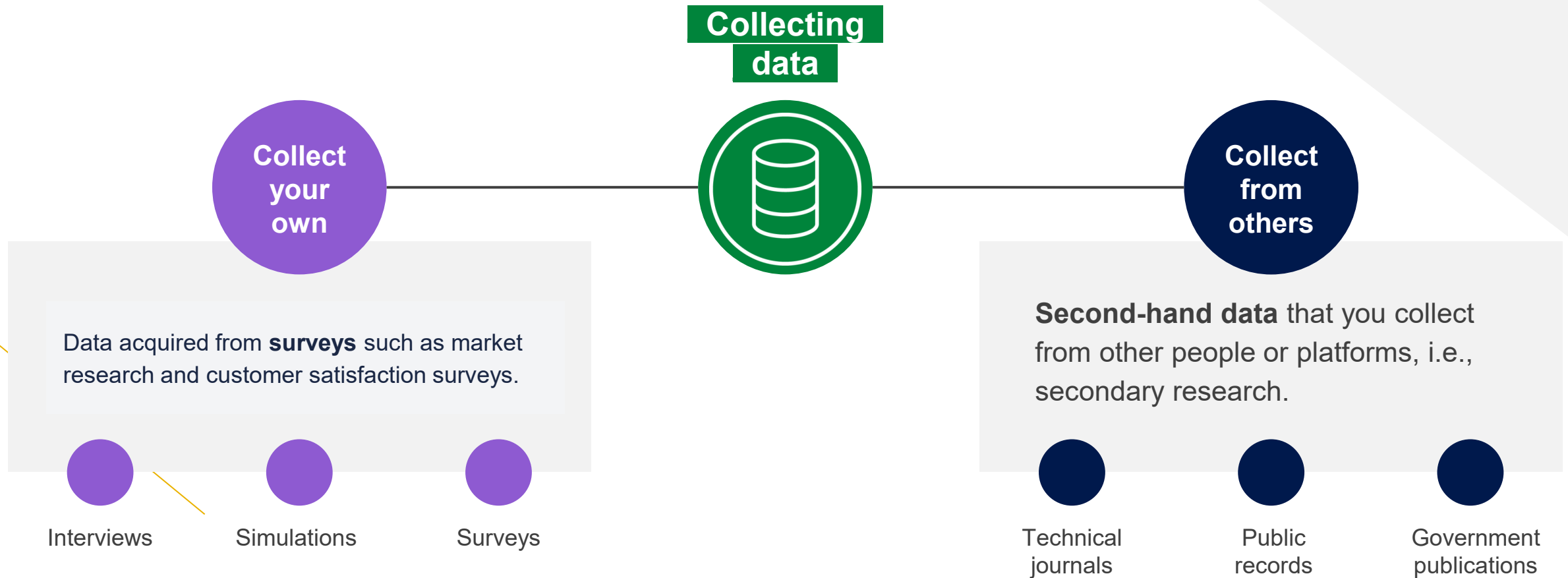
Data collection



Data collection includes **identifying and acquiring applicable data sources**, both internally and externally, which can help answer the problem statement.

We can use company data or open-source data, or collect our own data **depending on the nature** of our **problem** and the **analysis** we would like to do.

Let's Discover, Where to source your data?



Internal Vs External Data Sources

Readily available
data

Company
data
(Internal Data
Source)

Queried data from **databases** or **APIs** (Application Programming Interfaces) such as:



Sales
data



Financial
transactions



Customer
data



Open
source data
(External Data
Source)

Downloaded data from **open sources** and **cloud repositories** such as:



Medical data



Census
data



Weather
data

First-Party, Second-Party and Third-Party Data

Aspect	First-Party Data	Second-Party Data	Third-Party Data
Definition	Data collected directly from your customers.	Another business's first-party data shared with you.	Data collected by an external entity and sold to businesses.
Examples	Customer names, emails, purchase history.	Data from a partner with a similar audience.	Demographic data from a broad population.
Benefits	Highly accurate and relevant, provides detailed customer insights.	Access to new customer segments, insights from similar audiences.	Large datasets, helps in broad targeting and expanding reach.
Drawbacks	Limited to existing customers, not useful for reaching new audiences.	Potential privacy issues, data quality depends on the partner.	Variable accuracy, potential privacy concerns, less specific to your audience.

Real Time VS Batch Data

Aspect	Batch Data	Real-Time Data
Definition	Data collected and processed in large groups at scheduled intervals.	Data collected and processed continuously as events occur.
Examples	Daily sales reports, monthly financial statements.	Live stock market feeds, real-time social media analytics.
Benefits	Efficient for large volumes of data, lower processing cost.	Immediate insights, suitable for time-sensitive decisions.
Drawbacks	Delay in data availability, not suitable for time-sensitive insights.	Higher processing cost, more complex infrastructure.

Types Of Data Sources

- **Structured Data**
- **Unstructured Data**
- **Semi-Structured Data**

Types Of Data Sources

Aspect	Structured Data	Semi-structured Data	Unstructured Data
Definition	Data organized into rows and columns, easily searchable and analyzable.	Data that has some organizational properties but not a fixed schema.	Data that lacks a predefined format or structure.
Examples	Relational databases, spreadsheets.	JSON files, XML files, email.	Text documents, images, videos, social media posts.
Tools	SQL databases (MySQL, PostgreSQL), Excel.	NoSQL databases (MongoDB, Cassandra), data interchange formats.	Big data tools (Hadoop), text analytics tools, multimedia analysis tools.
Real-World Use Case	Financial transactions, inventory management.	Log files, configuration files, data integration tasks.	Customer feedback analysis, multimedia content management, social media monitoring.

Understand the data you need

Data is the link between the problem you are trying to solve and the intended solution. Ask the right questions using the 5W2H (what, why, where, when, who, how, how much) method to define the problem and understand the data that you need.

What data are you looking for?

- **Relevant** – data that pertain to the problem at hand and are in the required format.
- **Dependable** – data that are well structured, correct, and consistent.

NB: It **can't be 100% perfect**. It is all about finding a balance between the ideal and what is available.

What do you want to avoid?

- **Wrong data** – your analysis will be pointless (GIGO: garbage in, garbage out).
- **Too little data** – not enough to give you desired results.
- **Too much data** – cumbersome to work with and can cause time wastage.

Data in the real world

Even if we understand what we need from our data and that we can trust it, data in the real world is...

messy



difficult to work with

There are several ways to validate and convert real-world data into more usable and useful forms.



	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40.0	M	ATA	140	289.0	0	Normal	172.0	N	0.0	Up	0
1	NaN	F	NAP	160	180.0	0	Normal	156.0	N	1.0	Flat	1
2	37.0	M	ATA	130	283.0	0	ST	98.0	N	0.0	Up	0
3	48.0	F	ASY	138	214.0	0	Normal	NaN	Y	1.5	Flat	1
4	54.0	M	NAP	150	195.0	0	Normal	122.0	N	0.0	Up	0
5	39.0	M	NAP	120	339.0	0	Normal	170.0	N	0.0	Up	0
6	1000.0	F	ATA	130	237.0	0	Normal	170.0	N	0.0	Up	0
7	54.0	M	ATA	110	208.0	0	Normal	142.0	N	0.0	Up	0
8	37.0	M	ASY	140	NaN	0	Normal	130.0	Y	1.5	Flat	1
9	48.0	F	ATA	120	284.0	0	Normal	120.0	N	0.0	Up	0

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
19	36	M	ATA	120	267	0	Normal	160	N	3.0	Flat	1
20	43	F	TA	100	223	0	Normal	142	N	0.0	Up	0
21	43	F	TA	100	223	0	Normal	142	N	0.0	Up	0
22	43	F	TA	100	223	0	Normal	142	N	0.0	Up	0
23	44	M	ATA	150	288	0	Normal	150	Y	3.0	Flat	1
24	40	M	NAP	130	215	0	Normal	138	N	0.0	Up	0
25	36	M	NAP	130	209	0	Normal	178	N	0.0	Up	0

Dataset 1

	n	w	a
0	Ahmed	111	60
1	Seif	105	61
2	Nada	85	56

Dataset 2

	indvName	indvWeight	indvAge
0	Samy	105	36
1	Sara	60	25
2	Kareem	51	23

Dataset 3

	name	weight(kg)	age
0	Rana	118	55
1	Moaz	96	41
2	Noor	55	60

preprocess

preprocess

preprocess

	Name	Weight	Age
0	Ahmed	111	60
1	Seif	105	61
2	Nada	85	56
3	Samy	105	36
4	Sara	60	25
5	Kareem	51	23
6	Rana	118	55
7	Moaz	96	41
8	Noor	55	60

Real-world data is messy

We collect raw data in different ways and from a variety of sources; it is likely to be **messy** and **unclean**, which makes it pretty useless.

Unstructured data

Country KEN Year 2019 Proportion of seats
0.21776504298

Country South Africa Year 2019 Proportion
of seats 0.46347607053

Country Seychelles Year 2019 Proportion of
seats 0.21212121212

Country Sierra Leone Year 2019 Proportion
of seats 0.12328767123

Country Nigeria Year 2019 Proportion of
seats 0.03380281691

Inconsistencies

Missing values

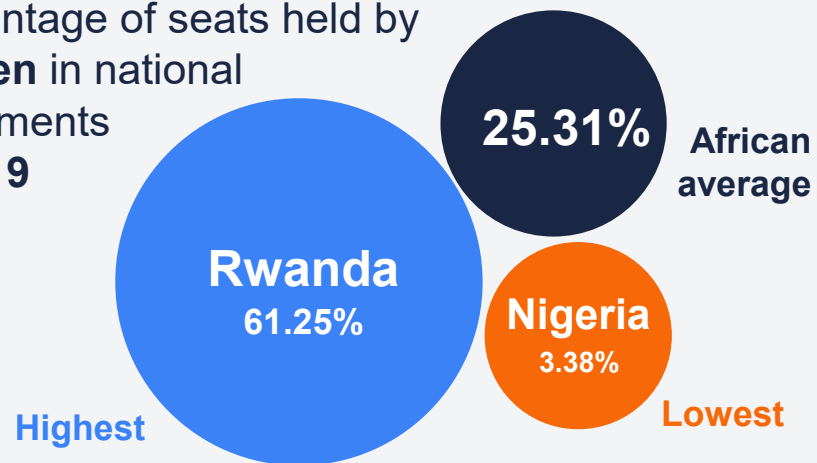
Erroneous values

Duplicates

Country	Year	Proportion of seats held by women in national parliaments
KEN	2019	0.21776504298
South Africa	2019	0.46347607053
Seychelles	2019	0.21212121212
Sierra Leone		0.12328767123
Nigeria	2019	0.03380281691
Rwanda	2019	o.6125
Nigeria	2019	0.03380281691

We clean this messy data to convert it into **useful, structured data** that can be used for **analysis**.

Percentage of seats held by **women** in national parliaments in **2019**



Country	Year	Proportion of seats held by women in national parliaments
Kenya	2019	0.21776504298
South Africa	2019	0.46347607053
Seychelles	2019	0.21212121212
Sierra Leone	2019	0.12328767123
Nigeria	2019	0.03380281691
Rwanda	2019	0.6125

Data validation

Data validation is the process of checking the accuracy and quality of data before use. We can put several checks in place that define the rules and constraints against which ingested data are compared.

■ Data type

It's of the **correct data type**, for example, numeric.

■ Format

It follows a **predefined format**, for example, a date format.

■ Length

It's of the **appropriate length**, for example, a phone number.

■ Consistent

It follows a consistent **logical order**, for example, check-in and check-out dates.

■ Range

It falls within a **specified range**, for example, dates within a month.

■ Uniqueness

Unique entries have **not** been **duplicated**, for example, IDs.

■ Presence

No entries have been left **blank**, especially for mandatory fields.

■ Look up

It conforms to a **set of acceptable values**, for example, days of the week.

Data cleaning



To address the errors identified **during data validation**, this is where **Data Cleaning** comes into play.

Data cleaning, also known as **data wrangling**, involves **transforming raw data into usable formats**.

We can use **several cleaning techniques to ensure** that our data are indeed **accurate** and of the required **quality**. If our data are inaccurate, so will our insights be.

Examples:

Using **spreadsheets** or a **programming language** to remove irrelevant observations, handle missing values, fix structural issues, etc.

Using **regular expressions** for pattern matching and replacing data.

Using **data visualisation tools** such as PowerBI or spreadsheets for identifying outliers and anomalies.

Exploratory data analysis



Exploratory data analysis (EDA) is an approach used to **summarise the main characteristics of a dataset** using aggregations, fundamental statistics, and visualisation techniques.

Before we can gather insights or build a model, we first need to **understand our data**. We can use **non-graphical methods**, such as descriptive statistics and correlation, or **graphical (visualisation) methods** to investigate our data.

Examples:

Descriptive statistics	Standard dev.
Aggregations	Count
Measures of central tendency	Mean
Measures of distribution	Kurtosis
Correlation	Pearson



Bar



Scatter



Density



Violin

Univariate and multivariate analyses

In EDA, we do either a univariate or multivariate analysis, depending on what we want to investigate.

Univariate analysis is the exploration of individual variables in a dataset, i.e. we only consider one variable at a time.

Non-graphical

We can use **descriptive statistics** such as the standard deviation, central tendency, and measures of distribution.

Graphical

We can use **visualisations** such as **histograms**, **density plots**, and **box plots** to understand the characteristics of a variable.

In a **multivariate analysis** we're more interested in the relationship between the different variables of our dataset.

Non-graphical

We use **correlation** to understand the strength and direction between variables.

Graphical

We can use **visualisations** such as **heatmaps**, **scatter plots**, and **pair plots** to investigate the relationship.

Gather insights



Gathering insights, also known as **data dissemination**, involves **gathering** and **reporting** the **insights** derived from the analysis.

Insights may be gathered in and reported to stakeholders through dashboards and reports that include text and data visualisations.

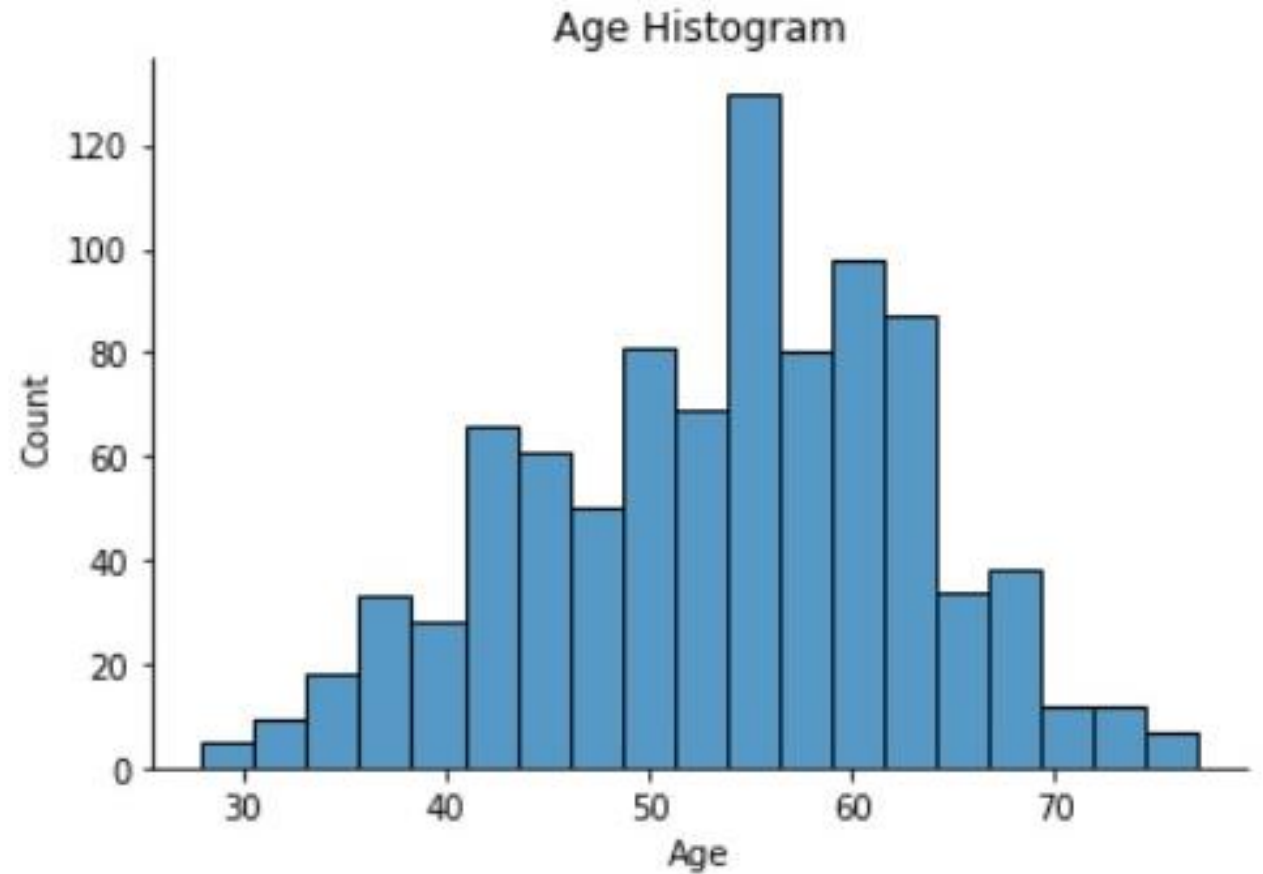
Examples:

Using **spreadsheets** or a **programming language** to summarise data and construct insights to form a report.

Using **data visualisation tools** such as PowerBI or spreadsheets to visualise and report the insights.

Univariate Analyses

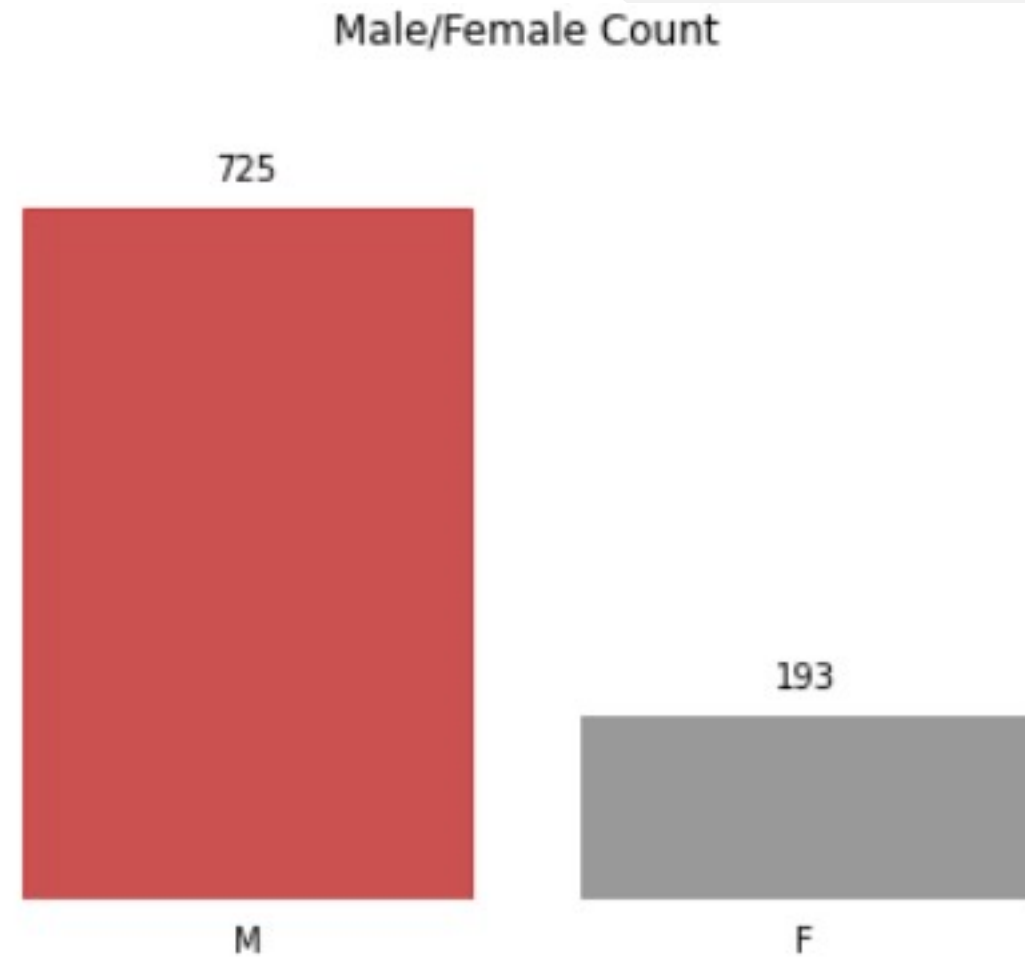
Age	
0	40
1	49
2	37
3	48
4	54
...	...
913	45
914	68
915	57
916	57
917	38



Univariate Analyses

NTI

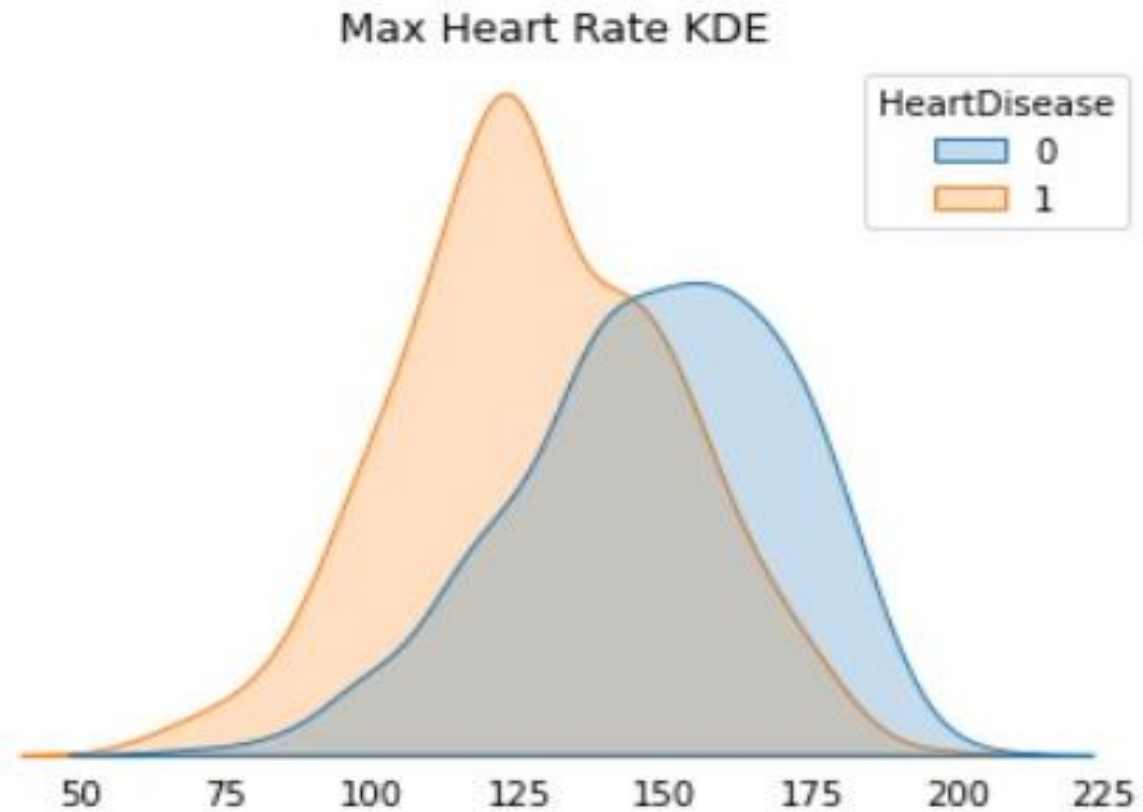
Sex	
0	M
1	F
2	M
3	F
4	M
...	...
913	M
914	M
915	M
916	F
917	M



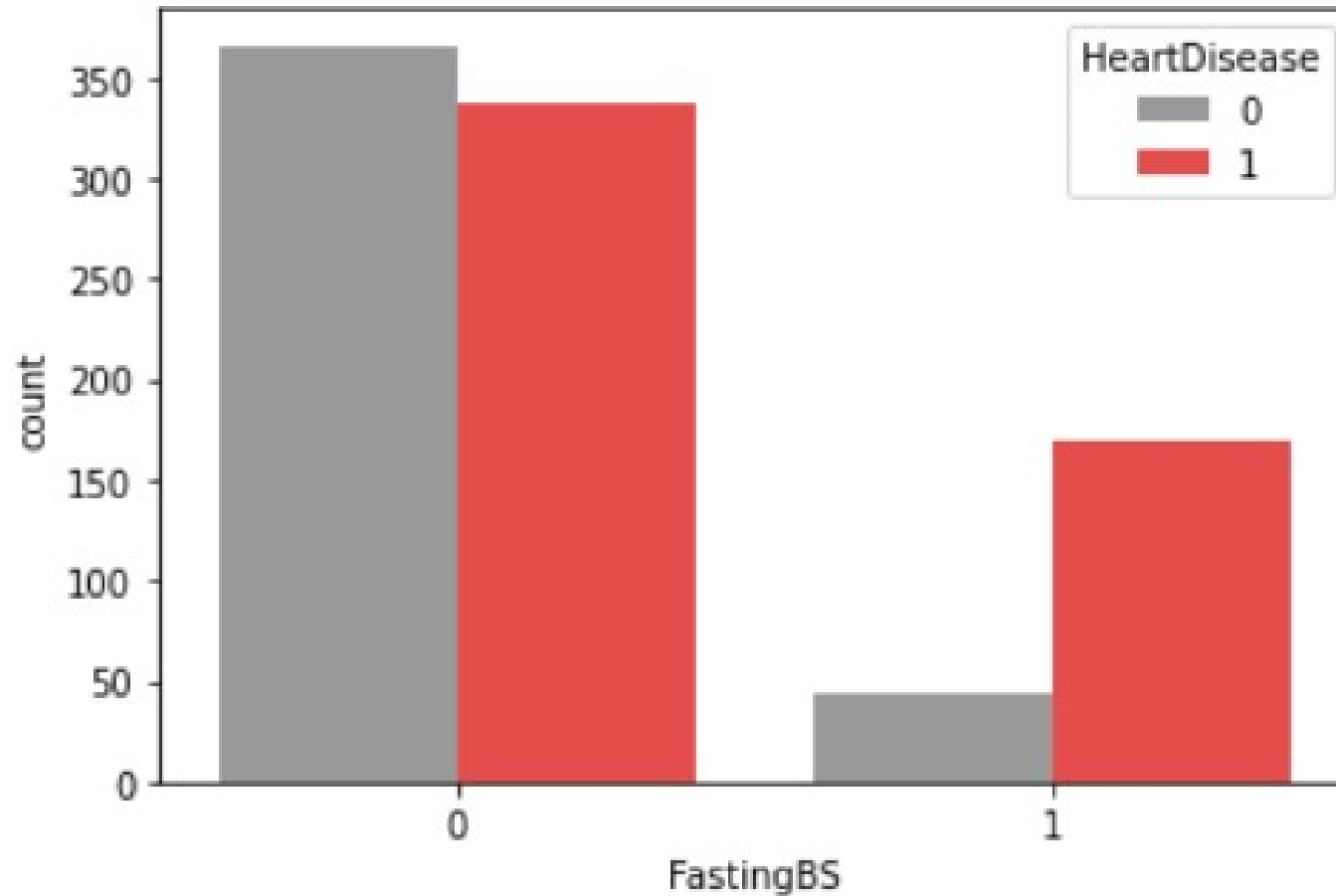
Multivariate Analyses



Multivariate Analyses



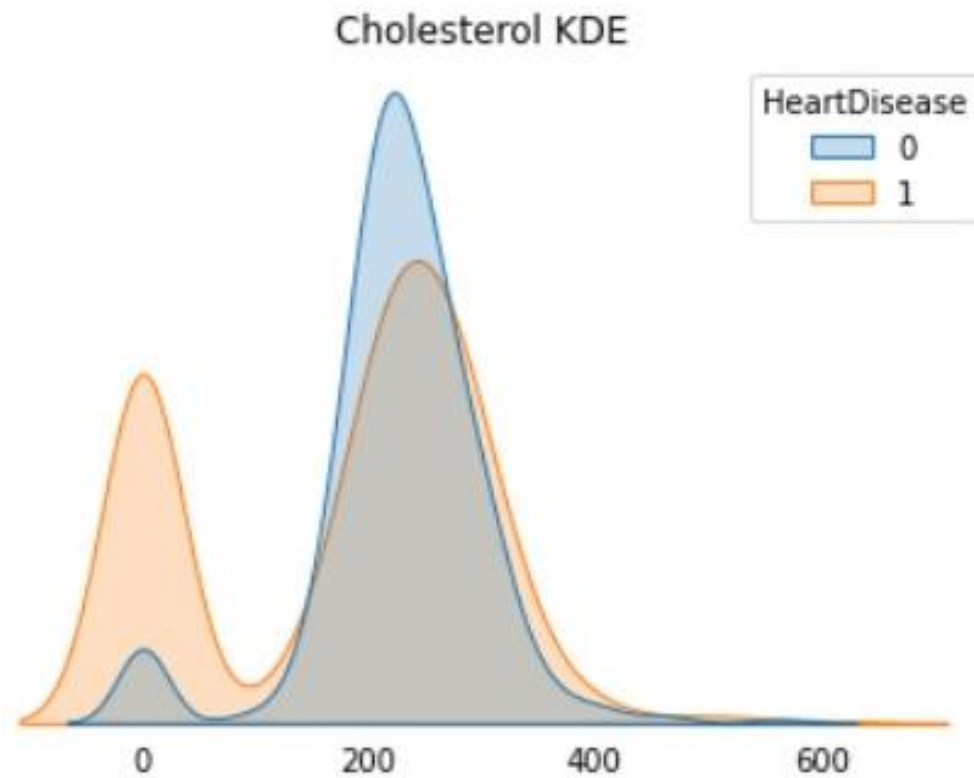
Multivariate Analyses



1: This indicates that the fasting blood sugar level is 120 mg/dL or higher.

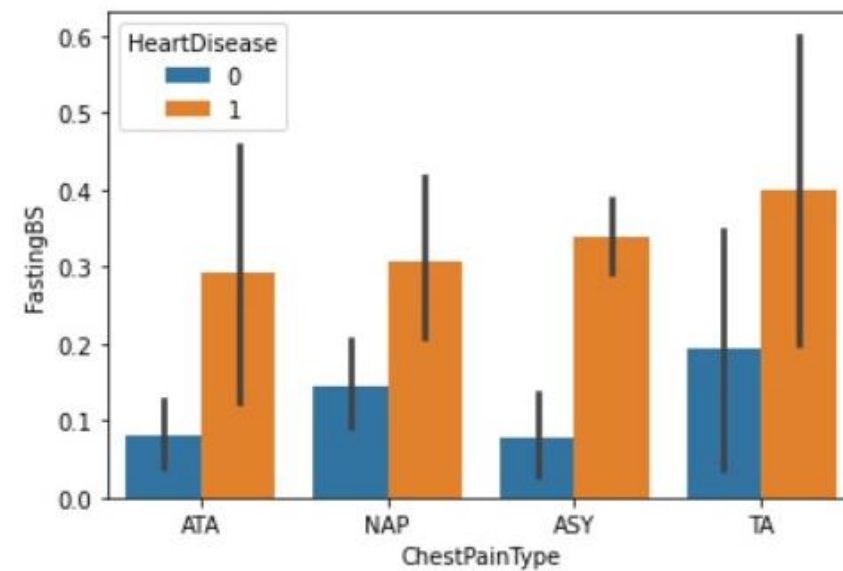
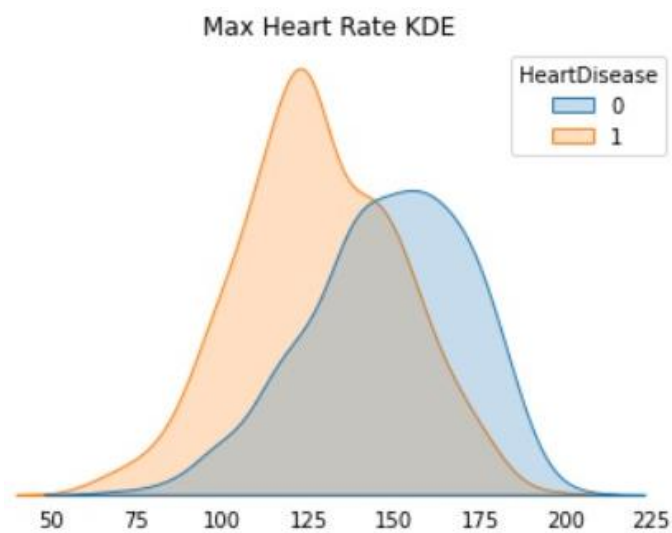
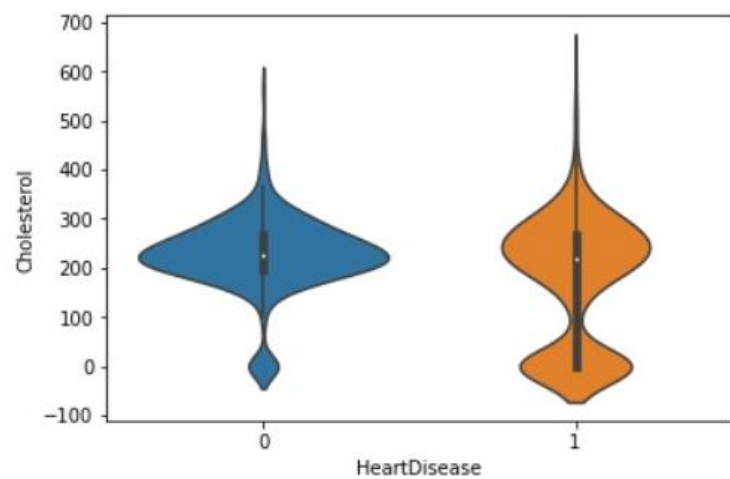
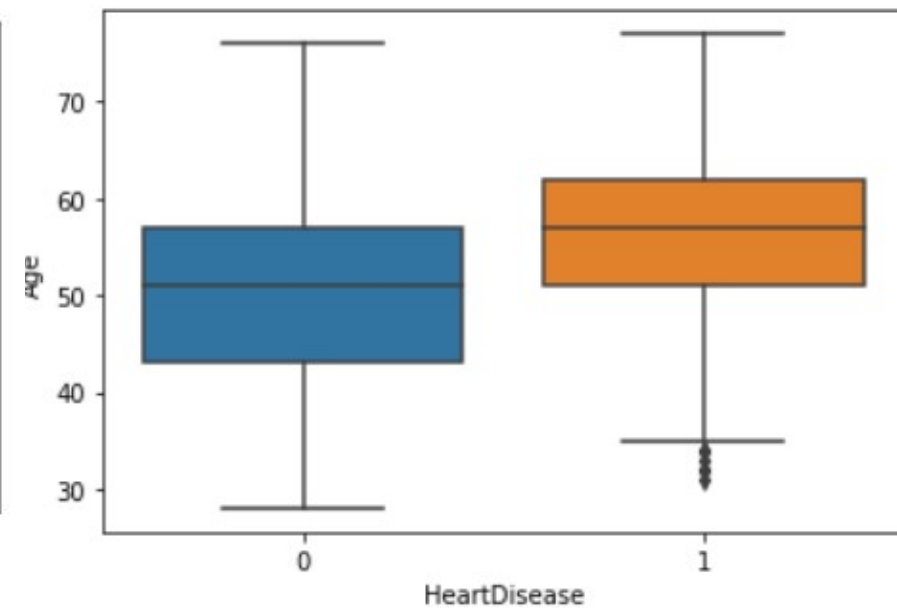
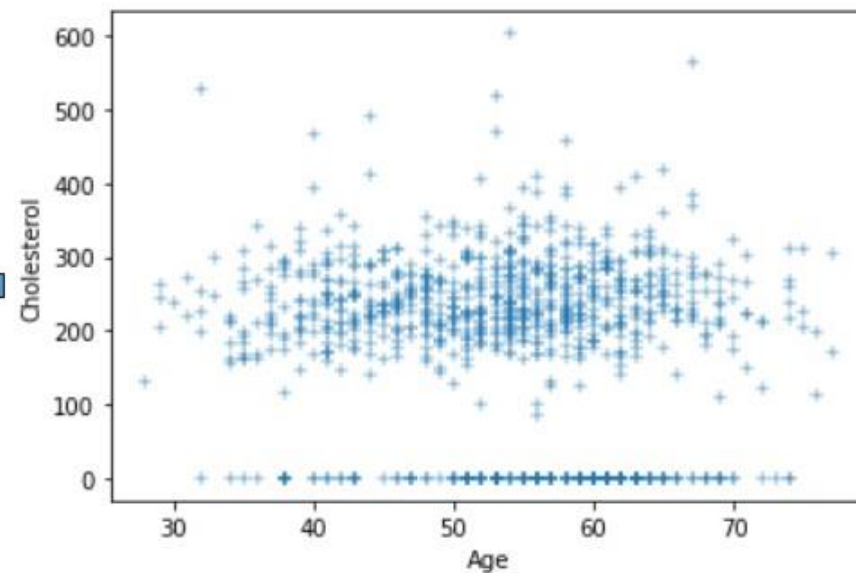
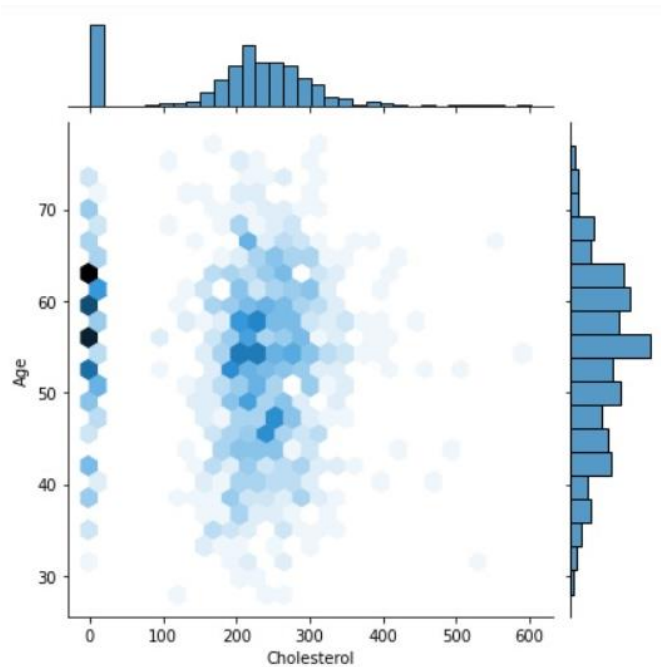
0: This signifies that the fasting blood sugar level is below 120 mg/dL, indicating a normal range without elevated glucose levels.

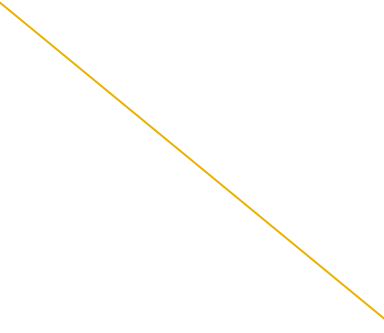
Multivariate Analyses



Insights

- Older people are more prone to heart diseases
- People with low heart rates are prone to heart diseases
- High Blood sugar indicates a possibility of heart disease
- Some patients have their Cholesterol level as 0, should address this





Data Analysis VS Data Analytics

Data Analysis VS Data Analytics

Data Analysis

- Focuses on examining and interpreting historical data to identify patterns and insights.
- Aims to understand what has happened by summarizing and visualizing data.
- Utilizes basic tools like Excel and SQL for querying and visualizing data.

Data Analytics

- Encompasses data analysis but also includes predictive and prescriptive techniques to inform decisions.
- Aims to predict what will happen and prescribe actions based on data
- Involves advanced tools and techniques, including statistical software and machine learning libraries.

Data Analysis VS Data Analytics

	Data Analysis	Data Analytics
Scope	Current Being	Current Being & Future
Goals	Describe & Diagnose	Predict & Prescript
Tools	Excel – SQL	Statistical & ML tools

Data Analysis Applications

- Medical Applications
- Business Analysis Applications
- Governmental Applications
- Digital Marketing Applications
- Business KPIs And Metrics

Type of analytics

The type of analytics we apply depends on our **goal** and prescribes our **approach** to the data analytics or data science process.

Descriptive

Hindsight

Used to describe **what** has happened in the **past**.

It's a summary of **historical data** that provides insights into patterns, trends, and relationships within the data

Examples: Dashboards and reports.

Diagnostic

Insight

Used to determine **why** something has happened in the **past**.

Helps organisations discover the causes of past outcomes..

Examples: Data mining and drill-down analysis.

Predictive

Foresight

Used to forecast **what** will happen in the **future**.

Uses statistical models and machine learning algorithms to identify patterns and trends in historical data to predict future outcomes.

Examples: Forecasting and risk modelling.

Prescriptive

Context

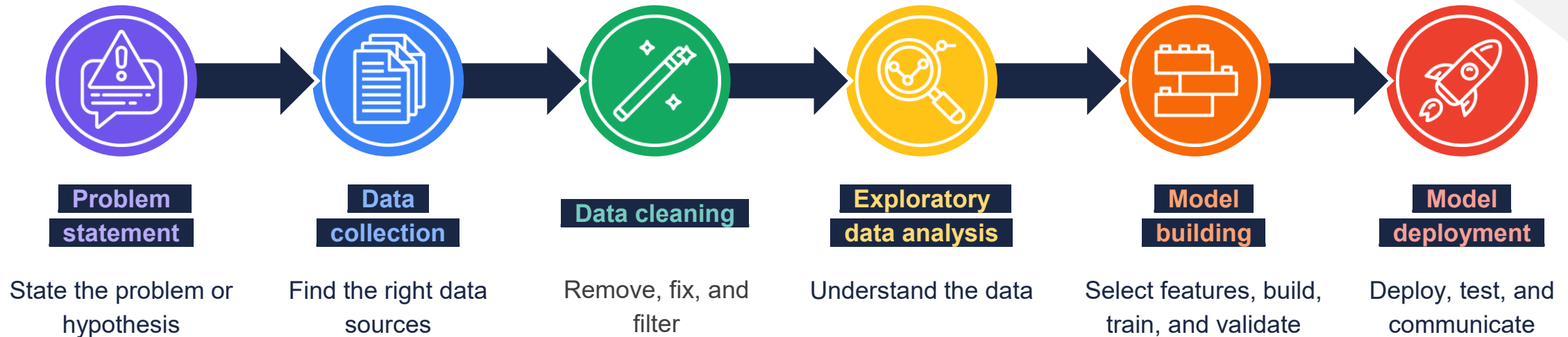
Used to recommend **What should we do in the future**, or the best **course of action** for a given situation.

Uses advanced algorithms and optimisation techniques to suggest the most optimal solution based on a variety of factors and constraints.

Examples: Optimisation

The data science process

The data science process is a systematic approach to **transforming a data problem** into a **data-driven solution**.



This approach to data science helps us to **discover meaningful** patterns, relationships, and trends and helps us develop **accurate** and **robust** models. **Various forms** of this process are used **across different data disciplines**, including data analytics, science, and engineering, under various names, such as OSEMN and CRISP-DM.

OSEMN Framework

Data Science Process



Obtain



Scrub



Explore



Model



iNterpret

Attributes of a data professional

Data professionals have many attributes that make them **effective** in **working with data**, **extracting insights**, and **deriving value** from data-driven approaches.



Analytical thinking

Tend to possess strong analytical skills and are capable of **breaking down complex problems into manageable components**. Can **identify patterns, trends, and insights** within data and use **critical thinking** to derive meaningful conclusions.



Technical proficiency

Tend to have a solid **understanding of data-related technologies, programming languages, and tools**. Proficient in data manipulation, analysis, and visualisation.



Curiosity and continuous learning

Tend to have a **natural curiosity** and a **passion** for **exploring and understanding data**. Eager to learn new techniques, tools, and methodologies to enhance their data analysis skills. Stay updated with the latest trends and advancements.

Attributes of a data professional



Attention to detail

Pays attention to detail when working with data. They ensure **data accuracy**, **completeness**, and **quality**. They are **meticulous** in data cleaning, pre-processing, and validation to minimise errors and biases that could impact analysis outcomes.



Collaboration and teamwork

Collaborate effectively with colleagues from different backgrounds, including stakeholders. They can **work together** to identify project goals, share insights, and contribute to data-driven decision-making.



Communication and storytelling

Skilled communicators who can effectively **convey complex data concepts**. Can translate data insights into clear, actionable recommendations and present findings through **effective use of visualisations** and **storytelling**.

While the specific combination and emphasis of attributes may vary depending on the role and organisation, these attributes are **commonly sought after** in data professionals, such as data analysts, data scientists, and data engineers.

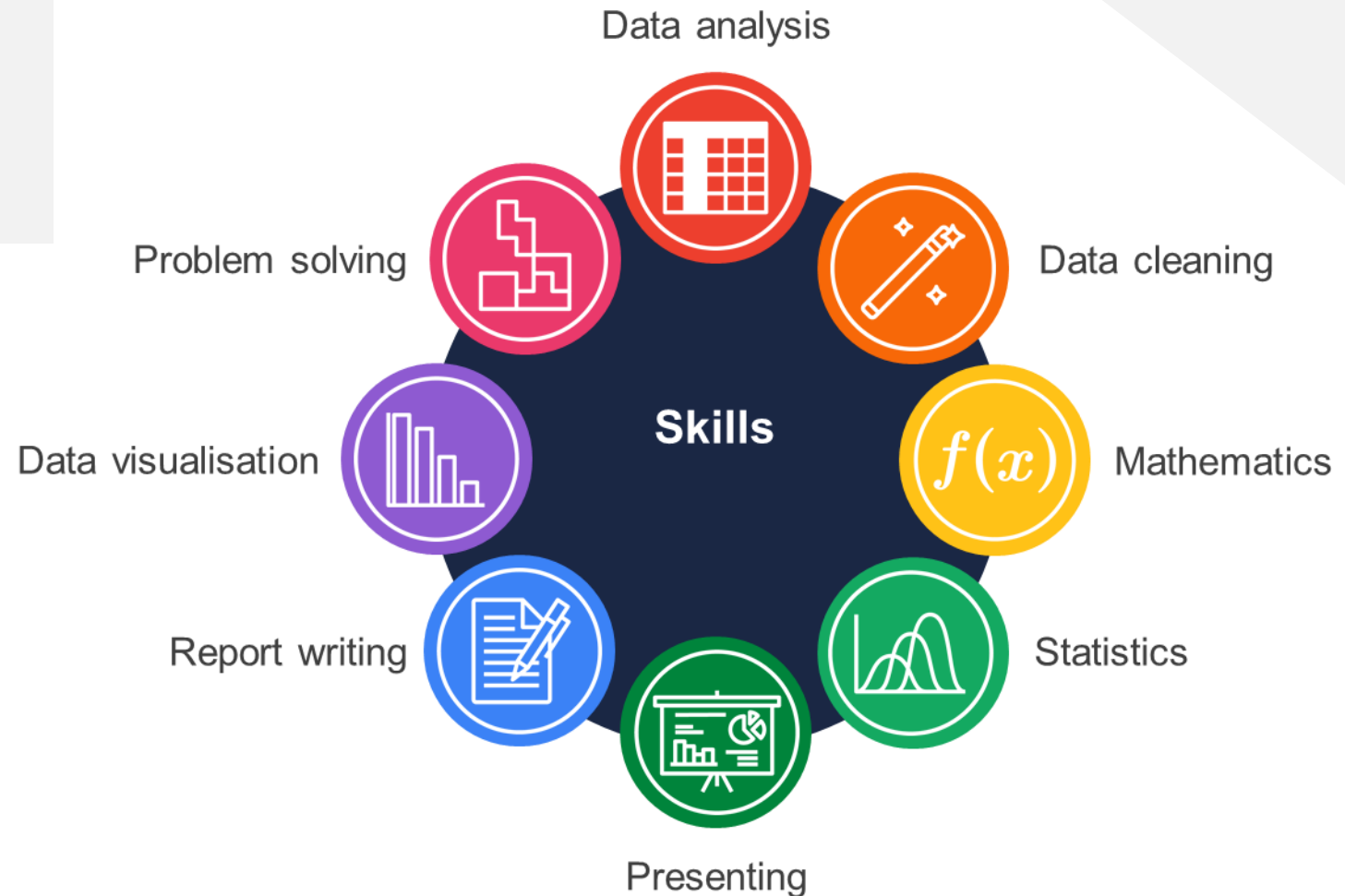
Since these **attributes are in high demand**, it is valuable for data professionals to **develop these attributes**.

Key data skills

Data has become an integral part of daily life and many of these skills are **critical** but **transferable** in many careers – we are all **data users**.

As data professionals, there are various skills we should have.

Depending on the specific role and responsibilities, **additional skills** such as machine learning, domain knowledge, database systems, and programming may be required.



Key data tools

Data professionals use several **different tools** to leverage their data skills and knowledge, depending on their role. Although we don't need to be proficient in all of them, it's important to know what the toolkit of a data professional could look like.



Spreadsheets:

Google Sheets, Microsoft Excel, Numbers.....



Database management systems:

MySQL, PostgreSQL, Oracle, Microsoft SQL Server, MongoDB, SQLite, Apache Cassandra



Data visualisation and dashboarding tools:

Microsoft Power BI, Tableau, QlikView



Programming languages:

SQL, Python, R



Integrated development environments (IDEs):

Visual Studio Code, Jupyter Notebook, PyCharm, RStudio



Version control systems:

Git, Mercurial, Subversion



Cloud computing platforms:

Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure



Machine learning frameworks:

scikit-learn (Python), TensorFlow, Keras

And many more.

Problem solving

Problem solving is a critical skill for all data users because it provides **structured**, **transferable**, and **logical** ways of approaching data problems.

Depending on the problem we need to solve, we use critical thinking “tools” such as **logic trees** and **flowcharts** to understand the problem and plan a solution.

When we understand the problem, we apply a **variety of skills and tools to solve** the problem. The tools we could use include:



Spreadsheets



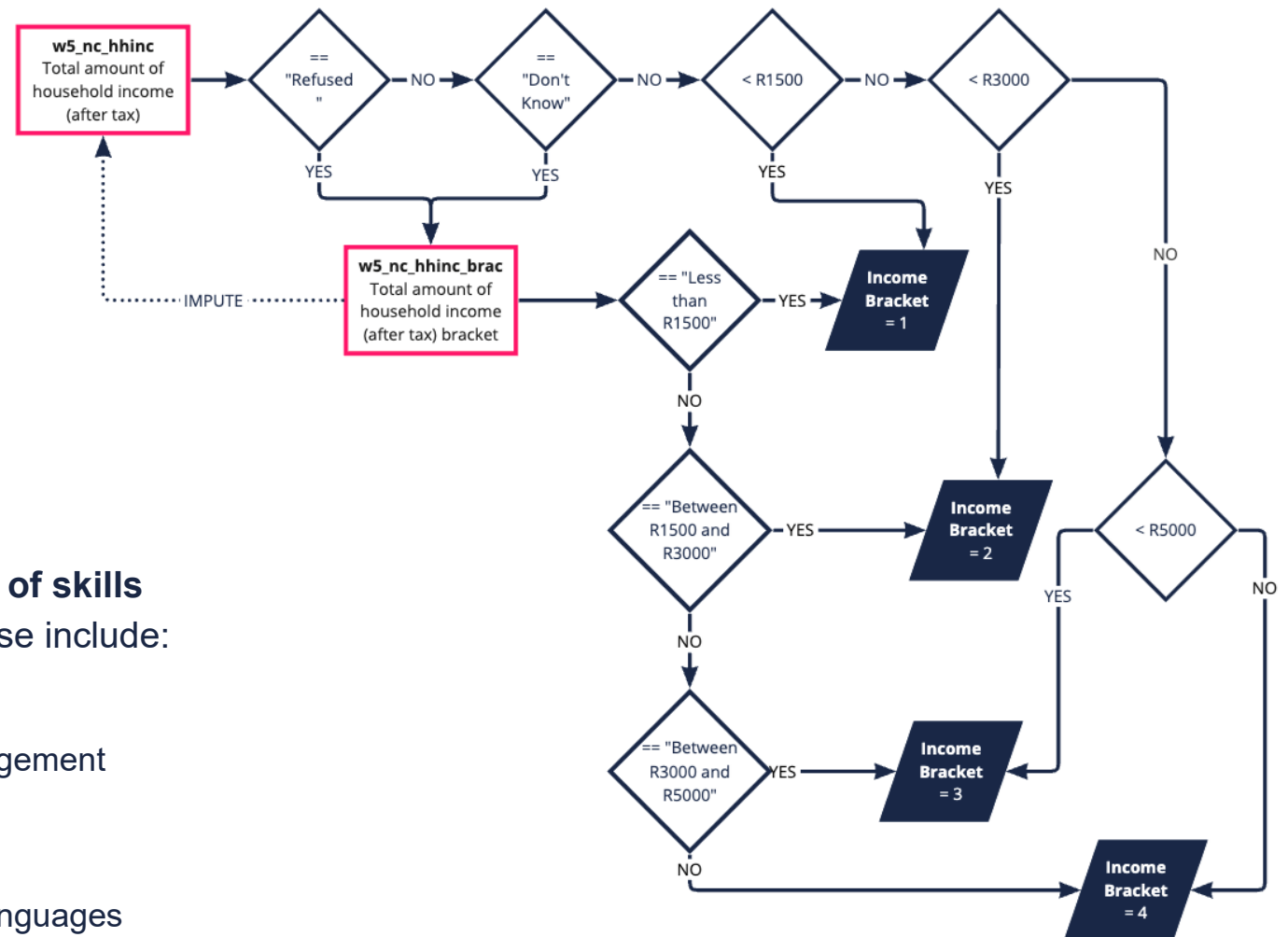
Database management systems



Data visualisation and dashboarding tools



Programming languages



Data preparation and analysis

Data preparation (cleaning) and **analysis** are critical skills for data professionals as they **enable effective decision-making, uncovering insights, identifying patterns, and detecting trends.**

The information extracted with these skills can **drive business strategy, optimisation, problem solving, and innovation.**

The tools we can use:



Spreadsheets



Database management systems

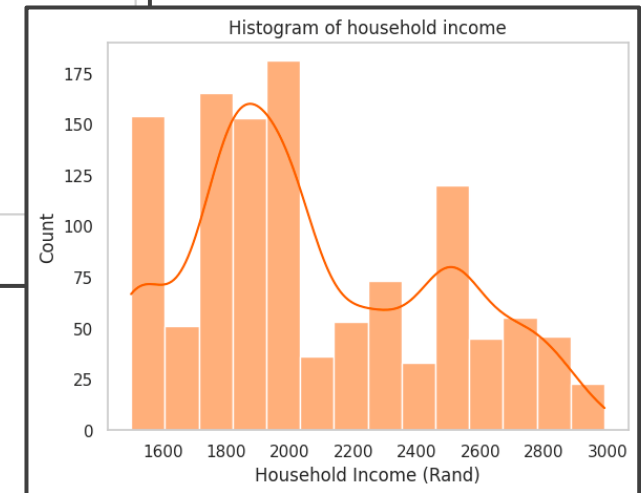
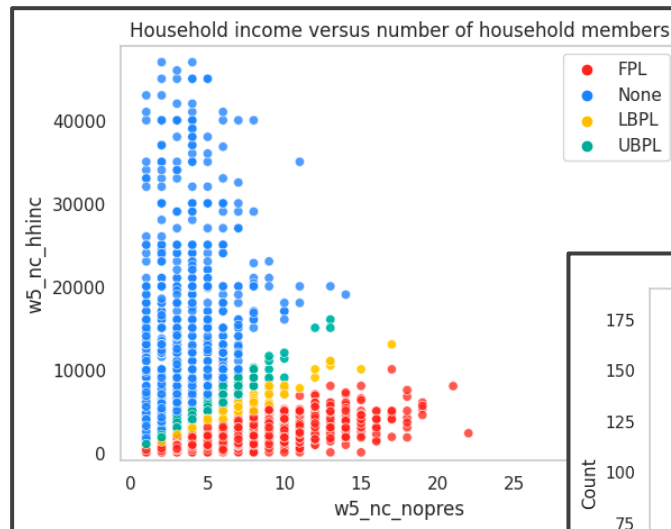


Data visualisation and dashboarding tools



Programming languages

Column	w5_nc_hhinc_brac	w5_nc_hhinc
Count non NaN	1561	5862
Count after drop "Refused" and "Don't Know" in w5_nc_hhinc_brac	1167	5468



Mathematics and statistics

Although **mathematics** and **statistics** may feel overwhelming to many, understanding the **fundamentals** of these fields makes **using data easier**.

Mathematics and statistics are **critical for data analysis, problem solving, and machine learning**.

We can apply this knowledge and skills across various tools, including:



Spreadsheets



Database management systems



Machine learning frameworks



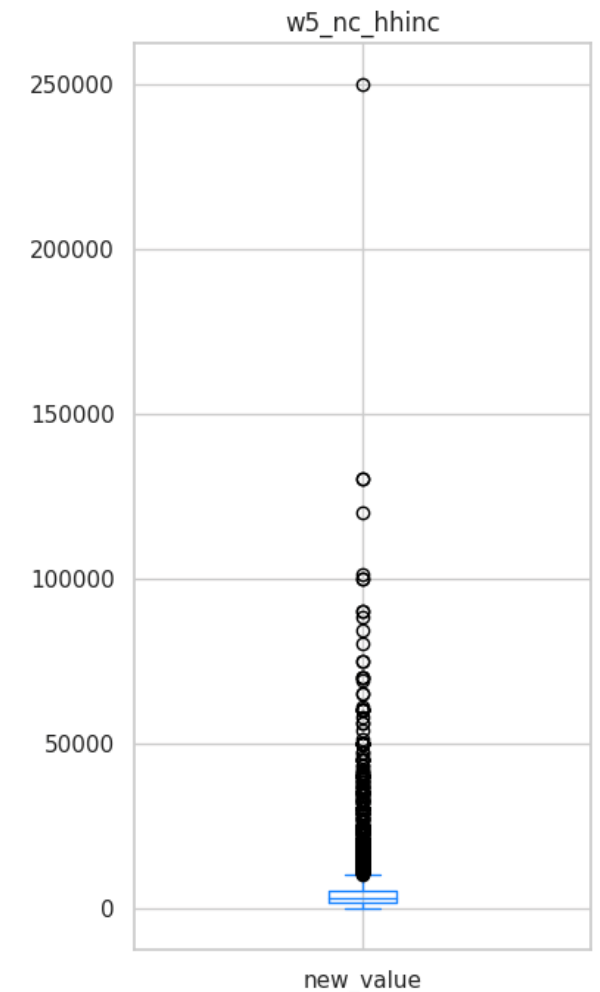
Programming languages

Column	w5_nc_hhinc
Count	4301.00
Mean	5653.48
Standard deviation	10057.09
Minimum	0.00
25%	1600.00
50%	3000.00
75%	5000.00
Maximum	250000.00

IQR = 3400

Higher outliers > R 10100

The number of outliers: 508 households (11.81%)



Communication and storytelling

Strong communication skills are essential for data practitioners to effectively share insights and findings to non-technical stakeholders.

The ability to **translate complex data concepts** into clear meaningful narratives helps organisations to make data-driven decisions.

We use **data visualisation, dashboards, reports, and presentations** to communicate data insights.

The tools we can use:



Spreadsheets



Data visualisation and dashboarding tools



Programming languages

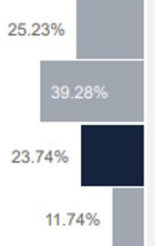
Unemployment and poverty in South Africa

INCOME PER CAPITA

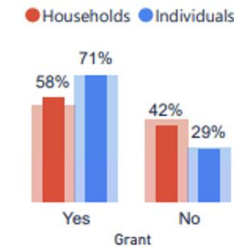


46.68

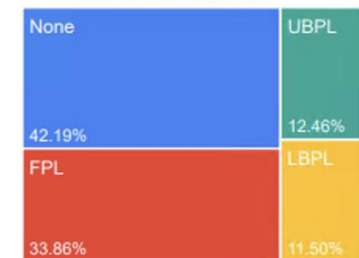
Mean Age 2017



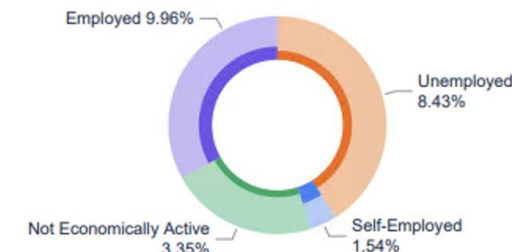
GRANTS AND EMPLOYED



POVERTY LEVEL



EMPLOYMENT



LFPR



UNEMPLOYMENT RATE



Databases, APIs, and web scraping

- **Database Key Concepts**
- **Relational Database**
- **Non-Relational Database**
- **APIs & Web Scraping**

Database Key Concepts

- Database Definition and Main Components
- DBMS and Data Models

Database Definition and Main Components

- A **Database** is an organized collection of data, typically stored and accessed electronically from a computer system. Databases are designed to manage, store, and retrieve large amounts of information efficiently, ensuring data integrity, consistency, and security.

Database Definition and Main Components

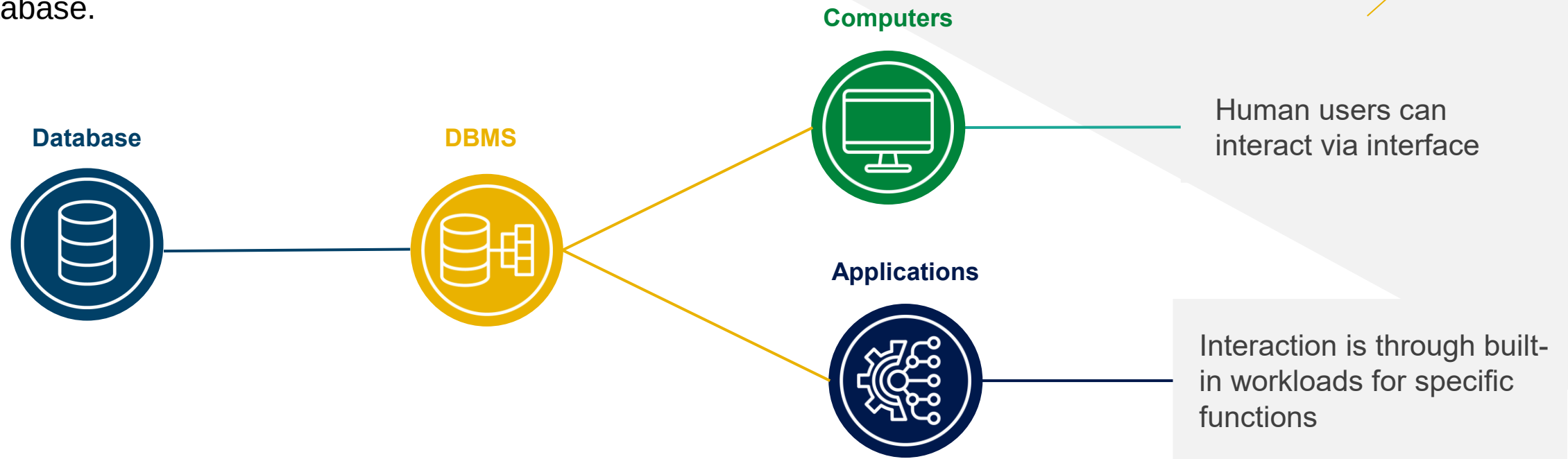
Key Component	Description
Database Schema	Structure defining the organization of data within the database, including tables, fields, and relationships.
Data	The actual information stored in the database, including structured, semi-structured, and unstructured data.
DBMS	Software that manages and interacts with the database, ensuring data integrity, consistency, and security.
Transactions	Sequences of operations performed as a single logical unit, adhering to ACID* properties for data integrity.
Query Language	Language used to perform operations on the data (e.g., SQL).
Views	Virtual tables created by querying one or more tables, providing a customized representation of data.
Stored Procedures and Functions	Predefined SQL code that performs specific tasks, enhancing performance and standardizing operations.

DBMS and Data Models

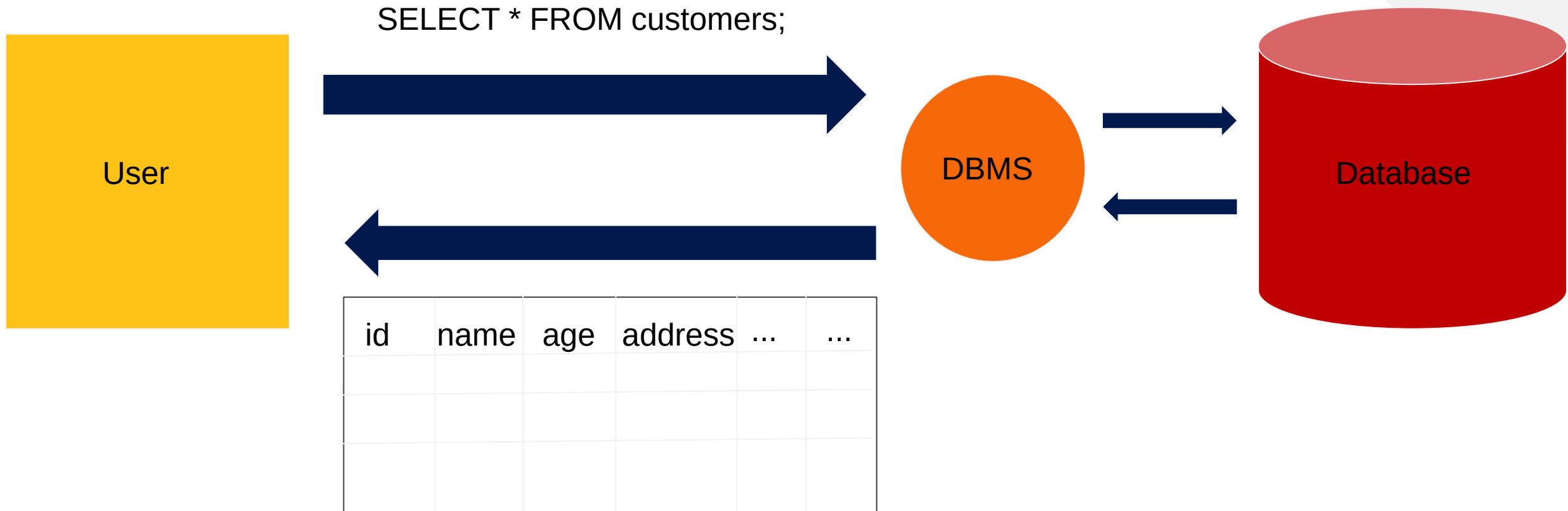
- A **Database Management System (DBMS)** is software that interacts with end-users, applications, and the database itself to capture and analyze data.
- The DBMS provides a systematic way to create, retrieve, update, and manage data. It ensures data integrity, consistency, security, and efficient data management.
- **Examples:** MySQL, PostgreSQL, Microsoft SQL Server, and MongoDB

What is a database management system?

A database management system (**DBMS**) is system software that allows users to **create, store, retrieve, and run queries on data** stored in a database. It acts as an **interface** between an end-user or an application and a database.



Databases



The purpose of database management systems

01. Data security

Incorporate security measures to **protect data** from **unauthorized access, manipulation, or breaches**.

02. Data manipulation and storage

Allow users to perform operations such as **inserting, deleting, and modifying** records. They manage the allocation of storage space, efficient **storage mechanisms**, and **data file management**.

03. Data backup and recovery

Allow for regular backups of the database which can be used to **restore data in case of system failures, data corruption, or other unexpected events**.

04. Data integrity

Implement various constraints, ensuring that data remains **accurate, consistent, and reliable**.

05. Data sharing and collaboration

Enable **multiple users to access** and work with the same data concurrently ensuring **data consistency**.

06. Data scalability

Handle **increasing volumes of data and growing user demands**. Scalability options include **partitioning or clustering**, to distribute and manage data across multiple servers.

Types of database management systems

Database management systems can be **categorised based on various criteria** such as the number of users or the data model. The most common types include:

01. Hierarchical DBMS

- Data are organised in a **tree-like structure**, where each parent node or record is linked to one or more child nodes, forming a **parent-child relationship**.
- For example: IBM's Information Management System (IMS)

02. Network DBMS

- Data are organised in a **graph-like structure**, where records are **connected by links** which represent their relationship. Child nodes or records can have **multiple parent** nodes or records, unlike the hierarchical model.
- For example: IDMS (Integrated Database Management System)

03. Relational DBMS

- Data are organised into **tables consisting of rows and columns**, where each table represents an entity or a relationship between entities.
- For example: Oracle, MySQL, Microsoft SQL Server, PostgreSQL, SQLite

ACID properties in RDBMSs

Relational database management systems should have **ACID** properties which ensure data validity and compliance. ACID is an acronym for **atomicity**, **consistency**, **isolation**, and **durability**.

A **transaction** is a **single unit** of work involving one or more operations, performed on a database with the aim of reading or modifying the data.

Transactions should follow **ACID** properties that guarantee the utmost data reliability and integrity.

For example, in the event of a power outage, the **absence of ACID properties** could mean that some of the **modifications made** to the database would **not be saved** causing inconvenience.

ACID properties of queries

A**Atomicity**

- Ensures that **all operations in a transaction or query** (to read, write, update, or delete data) are treated as a **single unit**.
- Meaning either the **entire query** is executed successfully if run, **or none of it is executed**.

C**Consistency**

- Ensures that transactions or queries only make changes to tables in **predefined, predictable ways**.
- This guarantees that errors or corruption in our data do not result in unintended consequences that **compromise the integrity of the database table**.

I**Isolation**

- Ensures that transactions or queries by multiple users on the same database **do not interfere with or affect one another**.
- Each query request can occur as though it were occurring one by one, even though they are simultaneously occurring.

D**Durability**

- Guarantees that modifications made to our data through successfully executed transactions will be **permanently saved**.
- This will remain the case in the event of a system failure.

RDBMSs support CRUD operations

Create

- Enables the creation of new records by inserting data into database tables.
- Usually achieved through query statements, e.g. `INSERT INTO` for SQL.

Read

- Enables the retrieval of data from database tables using SQL statements such as `SELECT`.
- The RDBMS executes the query and retrieves the matching data.

Update

- Allows modification of existing data in a database through SQL `UPDATE` statements.
- Verifies the constraints and applies the changes to the matching records, updating the data in the database accordingly.

Delete

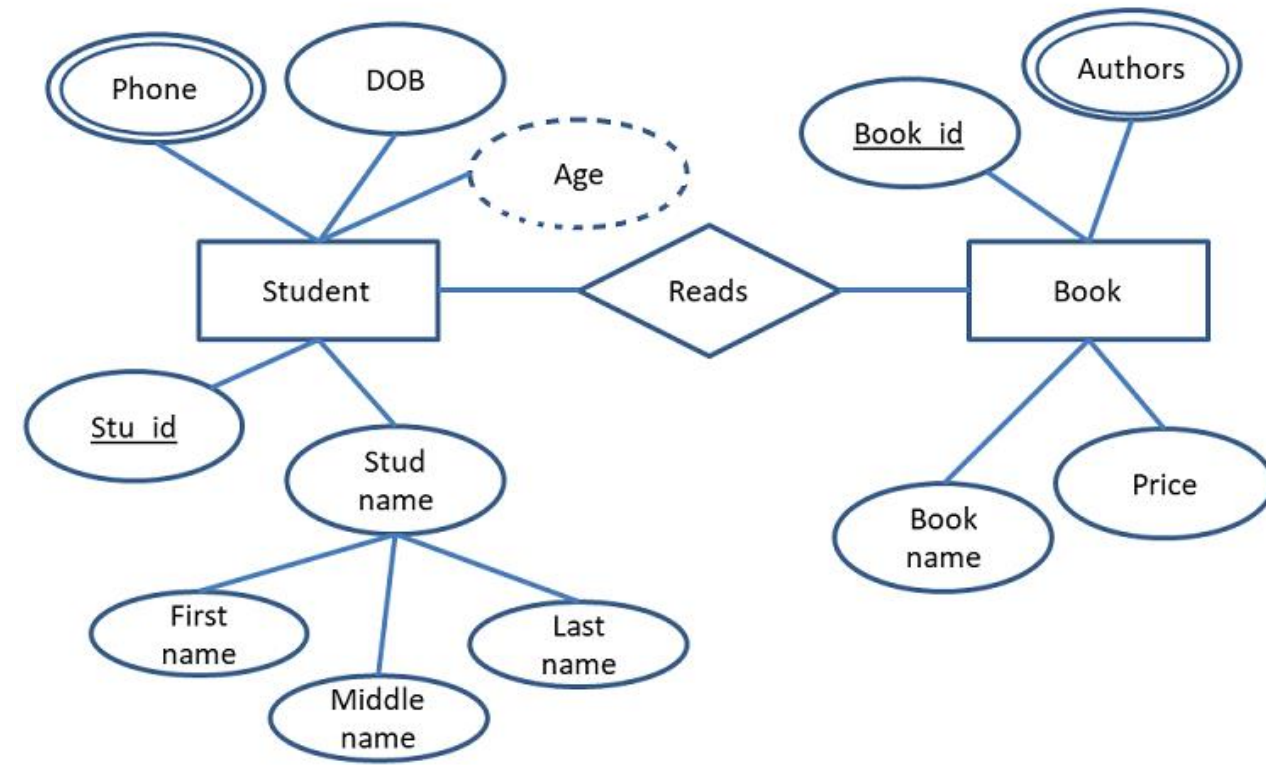
- Enables permanent removal of records from the database tables using SQL `DELETE` statements.
- Verifies constraints and deletes the matching records from the table, permanently removing them from the database.

Relational Database

- Entity Relationship Diagram (ERD)
- Relationships Degree
- Types of Relationships in ERD
- Normalization

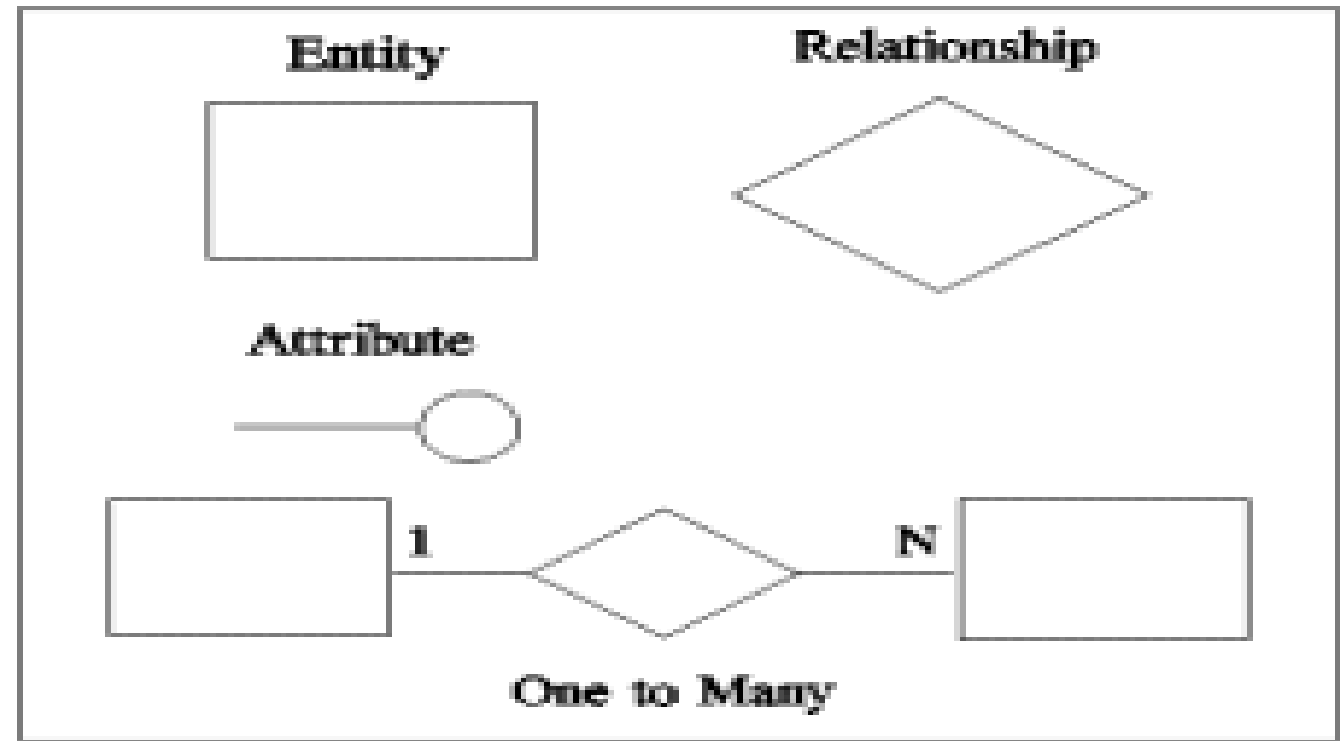
Entity Relationship Diagram (ERD)

is a graphical representation that depicts relationships among people, objects, places, concepts, or events in an information technology (IT) system. An ERD uses data modeling techniques that can help define business processes and serve as the foundation for a relational database.



ERD Notations

- Rectangles represent ENTITY CLASSES
- Circles represent ATTRIBUTES
- Diamonds represent RELATIONSHIPS
- Arcs - Arcs connect entities to relationships. Arcs are also used to connect attributes to entities. Some styles of entity-relationship diagrams use arrows and double arrows to indicate the one and the many in relationships.
- Some use forks etc.
- Underline - Key attributes of entities are underlined.



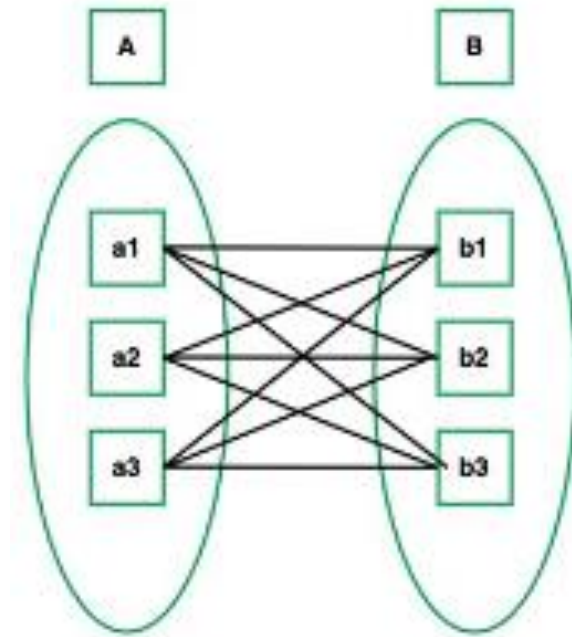
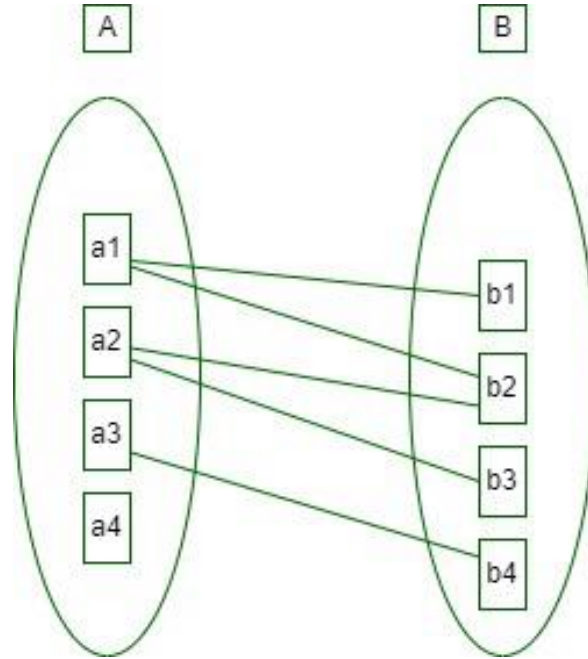
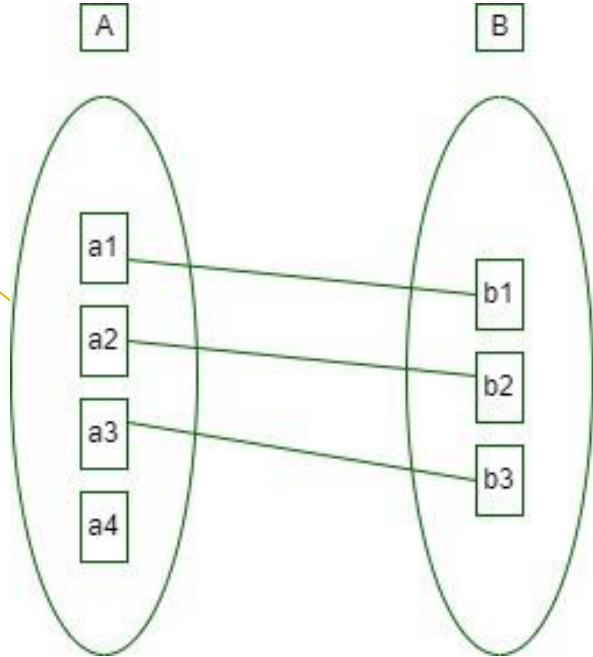
SUMMARY OF ERD NOTATION

<u>Symbol</u>	<u>Meaning</u>
	ENTITY TYPE
	WEAK ENTITY TYPE
	RELATIONSHIP TYPE
	IDENTIFYING RELATIONSHIP TYPE
	ATTRIBUTE
	KEY ATTRIBUTE
	MULTIVALUED ATTRIBUTE
	COMPOSITE ATTRIBUTE
	DERIVED ATTRIBUTE
	TOTAL PARTICIPATION OF E_2 IN R
	CARDINALITY RATIO 1:N FOR $E_1:E_2$ IN R

Types of Relationships in ERD

- One to One Relationship
- One to Many Relationship
- Many to many Relationship

Types of Relationships in ERD



Normalization

- What is Normalization
- Functional Dependency
- Rules of 1st Normal Form
- Rules of 2nd Normal Form
- Rules of 3rd Normal Form

What is Normalization

- Normalization is a process in database design that organizes tables and attributes to reduce redundancy and dependency.
- It typically involves multiple phases, commonly referred to as Normal Forms (NF).

Functional Dependency

- Functional dependency describes the relationship between attributes in a relational database.
- It indicates that the value of one attribute (or a group of attributes) determines the value of another attribute.

Normalization Roles

Normal Form	Key Characteristics	Primary Objective
1NF (First Normal Form)	- Eliminates repeating groups. - Ensures atomicity of columns (each column has a single value). - Identifies a primary key for each row.	To eliminate redundancy and ensure each column contains only atomic values.
2NF (Second Normal Form)	- Removes partial dependencies. - Ensures non-key attributes are fully functionally dependent on the primary key. - Non-key attributes dependent on part of the primary key are moved to separate tables.	To remove redundancy and ensure all non-key attributes are dependent only on the whole primary key.
3NF (Third Normal Form)	- Eliminates transitive dependencies. - Ensures non-key attributes are not dependent on other non-key attributes. - Non-key attributes dependent on other non-key attributes are moved to separate tables.	To further reduce redundancy and ensure data integrity by removing indirect relationships between non-key attributes.

APIs & Web Scraping

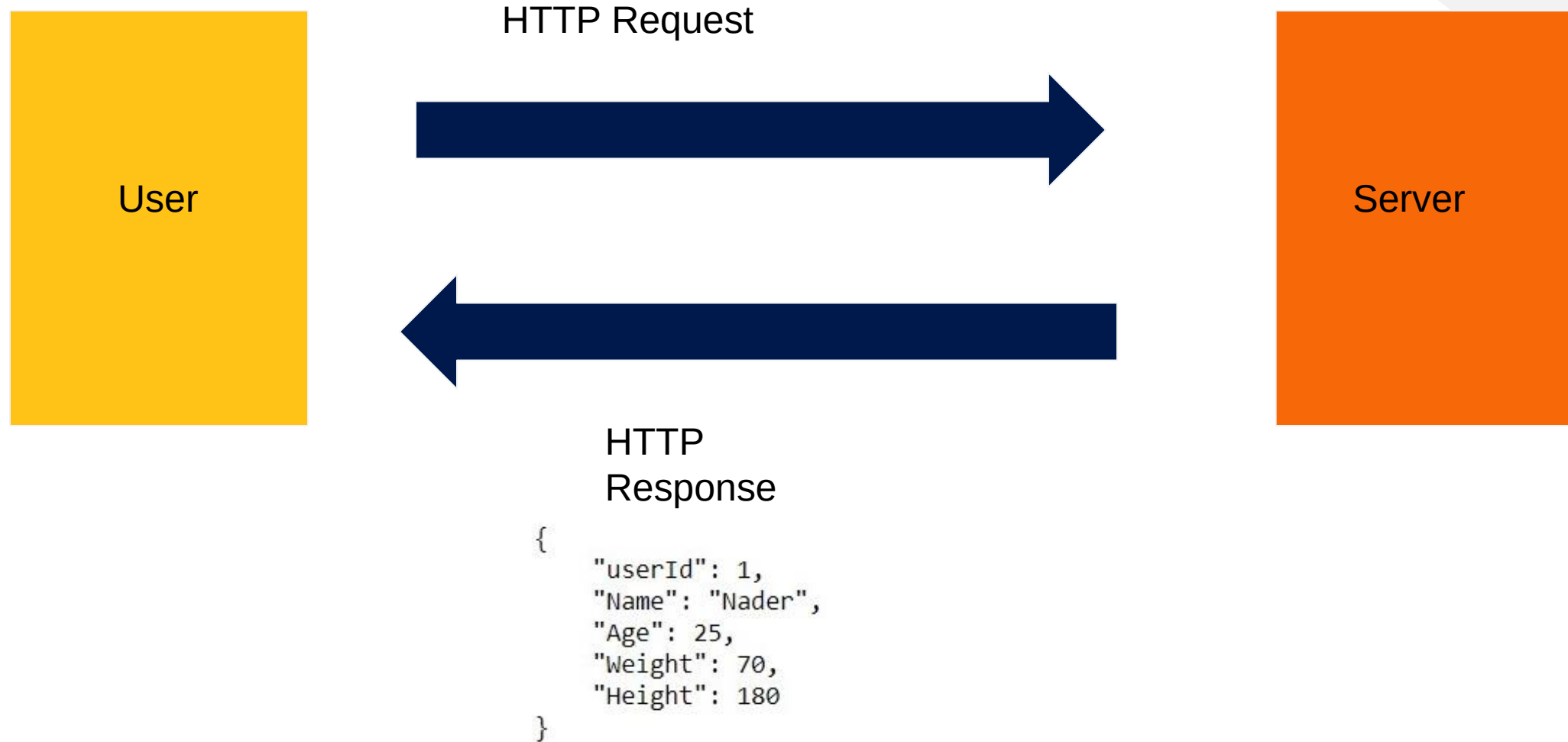
- APIs Definition
- APIs Examples: Twitter API, Google Maps API.
- How to connect to an API and retrieve data.
- Python Application (Web Scraping)
- Ethical and legal considerations.

APIs

- An API, or Application Programming Interface, is a set of rules and protocols for building and interacting with software applications.
- It defines the methods and data formats that applications can use to communicate with each other.

Web APIs

NTI



Types of APIs

- Web APIs:
 - REST (Representational State Transfer)
 - SOAP (Simple Object Access Protocol)
- Library-Based APIs:
 - Sets of routines, protocols, and tools for building software and applications.
 - For example, the standard libraries in languages like Java, C++, or Python.

Web Scrapping



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```
Welcome Elements Console Sources » + 1 17 62
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Styles Computed Layout Event Listeners DOM Breakpoints Properties Accessibility
```

Online Datasets

≡

kaggle

+

Create

🏠

Home

🏆

Competitions

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Datasets

⏏

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Datasets

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See All



Customer Personality Analysis

Akash Patel · Updated 2 months ago

Usability 9.7 · 63 kB

1 Task · 1 File (CSV)

670





2021 Olympics in Tokyo

Arjun Prasad Sarkhel · Updated 3 mont...

Usability 8.2 · 356 kB

1 Task · 5 Files (other)

911





Heart Failure Prediction Dataset

fedesoriano · Updated 2 months ago

Usability 10.0 · 9 kB

1 Task · 1 File (CSV)

391





Turkey Covid 19 Vaccination Data

Omer Colakoglu · Updated a month ago

Usability 4.1 · 13 kB

1 File (other)

287



🕒

Recently Viewed Datasets

See All









Questions





Thank You.



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